VET2EMPLOY RAG SERVICE | BRIEFING 1 OF 5 ON RAG

Retrieval Augmented Generation

An AI model that **looks things up in a trusted library** before it answers, and **shows the page** each answer came from.

Think of the difference between sitting an exam from memory and sitting it with the right textbook open on the desk.

R — **Retrieval**
Find the few passages that answer the question.

A — **Augmented**
Add those passages to the question the model receives.

G — **Generation**
Write the answer from those passages, with citations.

RAG EXPLAINED | 2 OF 5

Why a Model Needs a Library

CLOSED BOOK: THE MODEL ALONE

Question: “When is the Data Quality Log due?”

↓

Model answers from memory: general patterns learned months ago, nothing about Vet2Employ

↓

“It is due Friday at noon.” Fluent, confident and invented. No source.

OPEN BOOK: THE MODEL WITH RAG

Question: “When is the Data Quality Log due?”

↓

Search the Work Brief first: the deliverables table, section 7.7, is found

↓

“D2 is due Tuesday at 17:00 as an XLSX file.” Source: Work Brief, pages 31 to 32.

Hallucination is fluent text that is not true. It comes from how language models work, predicting likely words, so it cannot be patched away. It has to be designed around.

How RAG Works in Four Moves

ONCE, BEFORE ANY QUESTION

EVERY TIME A QUESTION ARRIVES

1

Index

Cut documents into passages, turn each into a numeric fingerprint, store them in a searchable library.

2

Retrieve

Turn the question into a fingerprint and fetch the passages that sit closest to it.

3

Augment

Place those passages into the instructions the model receives, with rules on how to use them.

4

Generate

The model writes the answer from the passages and names which passage supports each point.

Library analogy: cataloguing the books (1), the librarian fetching the right pages (2), handing them to the writer (3), the writer quoting them (4).

Inside the Machinery: Passages, Fingerprints, Distance

A **Chunk**

Split each document into passages of a few hundred words, each tagged with where it came from.

B **Embed**

A model turns each passage into a list of 1,536 numbers that captures its meaning.

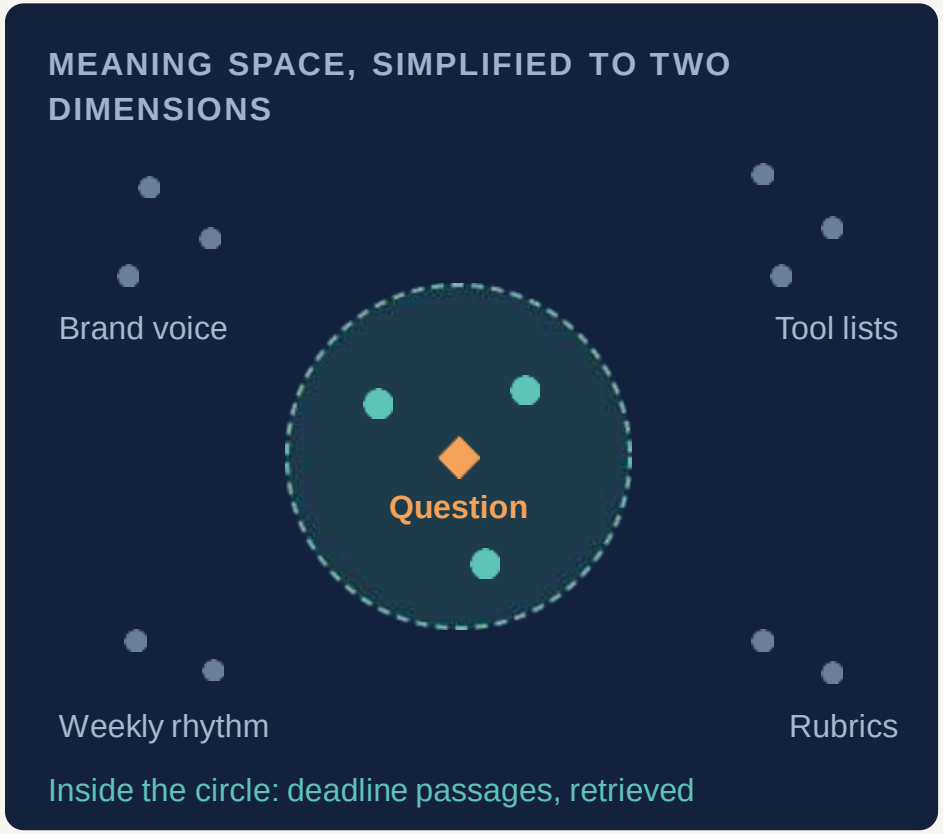
```
[ 0.12, −0.48, 0.91, 0.07, … ]
```

C **Compare**

Passages whose numbers point the same way as the question’s are close in meaning. The closest are retrieved.

D **Assemble**

Rules, numbered passages and the question become one prompt for the model.



RAG EXPLAINED | 5 OF 5

What RAG Gives a Business, and Where It Stops

Accuracy
Answers come from your documents, not general memory.

Freshness
Update a file and the next answer uses it. No retraining.

Traceability
Every point can name the passage it came from.

Control
You decide what the model is allowed to know.

Cost
Cheaper than training a custom model, and reversible.

WHERE BASIC RAG STOPS

Retrieval improves the odds. It does not prove the answer.

The search can miss. One question may need a deadline, a rubric and a client fact from three places.

The model can drift. It may blend a passage with memory, or fill a gap with a plausible detail.

Nothing checks the result. Basic RAG sends the answer straight to the reader.

The rest of this deck shows how Vet2Employ closes all three gaps.

PHEXARA LIMITED | VET2EMPLOY PLATFORM

The Vet2Employ RAG Service

Five architectures analysed, one selected, and the codebase explained step by step for technical and non technical stakeholders

Python and FastAPI

PostgreSQL with pgvector

OpenRouter

BEFORE WE START

How to Read This Deck

IF YOU ARE NOT TECHNICAL

Follow the titles and the boxes marked “In plain terms”.

Each part opens with what it means for candidates, employers and the business. Skip the code slides without losing the story.

IF YOU ARE TECHNICAL

Look for the panels marked “Under the hood”.

They name the module, the function and the exact numbers: weights, thresholds, budgets and test results.

DIAGRAM COLOUR KEY, USED ON EVERY ARCHITECTURE SLIDE

Retrieval | Model and generation | Verification and safety | Data and storage | Platform and outside

AGENDA

Ten Parts, From First Principles to Deployment

01	RAG explained	1 to 5
02	The problem and the goal	6 to 16
03	The job simulation market	17 to 21
04	Five RAG architectures analysed	22 to 59
05	Comparison and decision	60 to 66
06	The selected architecture in depth	67 to 95
07	The knowledge base	96 to 115
08	The codebase, step by step	116 to 165
09	Quality, integration and operations	166 to 184
10	Decisions, roadmap and close	185 to 194

EXECUTIVE SUMMARY

The AI Manager Now Speaks Only From Verified Evidence

THE PROBLEM

A realistic simulation cannot tolerate invented facts.

A language model alone invents deadlines, colleagues and tools. Each invention breaks the simulation and weakens the Verified Profile an employer relies on.

THE SOLUTION

A planned and verified RAG service between the platform and OpenRouter.

It retrieves exactly the evidence a task needs, lets the model write only from it, and checks every claim before release.

THE OUTCOME

Tasks that are realistic, consistent and auditable.

Every statement cites the Work Brief or real UK employer research. When evidence is missing, the service declines rather than guesses.

Selected after comparing five architectures against six weighted requirements: it scored 4.65 out of 5, against 3.00 for the next best option.

AT A GLANCE

The Service in Six Numbers

27

source files ingested with zero failures

3,194

searchable passages, each tagged and citable

521

researched UK job roles across four disciplines

48 / 48

programme activities detected: 12 in each of 4 tracks

7

independent layers of hallucination control

96

automated tests passing, including real database runs

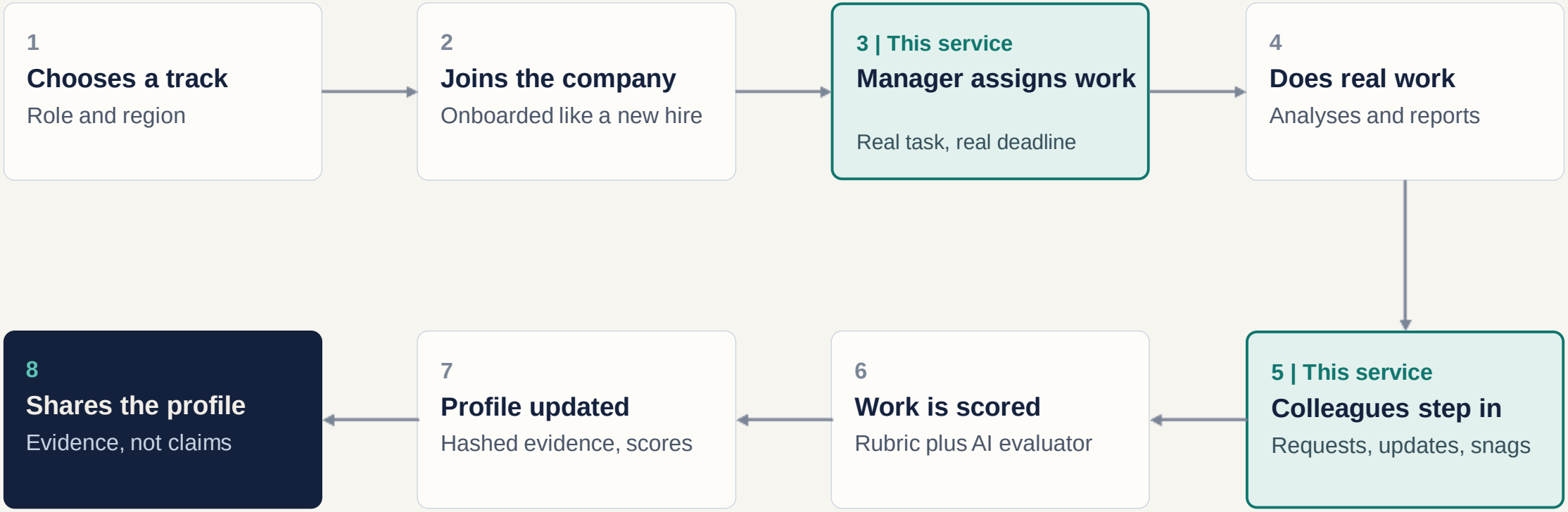
02

The Problem and the Goal

What Vet2Employ promises candidates and employers, why an ungrounded AI Manager puts that promise at risk, and the six requirements any solution had to meet.

THE PRODUCT

A Candidate’s Journey Through Vet2Employ



Teal outline: moments where the AI Manager or a colleague persona speaks. Every one of them is grounded by this service. Source: Vet2Employ Plain English Report, Part 3.

THE STANDARD WE ARE HELD TO

“The AI manager is not a chatbot. It is a carefully designed professional persona that has been tested and refined until real professionals in that field say it behaves the way an actual manager in that role would behave. **This realism is the most critical part of the product.**”

Vet2Employ Plain English Report, Part 3: How It Works

THE RISK

Five Ways an Ungrounded Manager Gets It Wrong

Kind of error	What a model alone might write	What the evidence actually says
Invented deadline	“Send me the quality log by Friday noon.”	D2 Data Quality Log is due Tue 17:00
Invented colleague	“Sarah in Finance will send you the files.”	The sponsor is Priya Raman, Head of Commercial Insight
Invented tool	“Load everything into Snowflake first.”	Excel, a SQL environment and Power BI or Tableau
Invented file name	“Save it as DQ_Final.xlsx.”	V2E_DA_[Surname]_D2_DataQualityLog_v1.xlsx
Wrong client fact	“Kestrel Lane runs 50 stores nationwide.”	38 stores across the North West and the Midlands

Left column: illustrative examples of typical model errors. Right column: facts quoted from the Vet2Employ Week One Work Brief, Data Analyst track.

THE COST OF ONE ERROR

How One Invented Detail Reaches a Hiring Decision

1

A detail is invented

The manager names a deadline the brief never set.

2

The candidate acts on it

They plan their week around the wrong time.

3

The scoring disagrees

The rubric follows the brief, so the candidate is marked late.

4

The profile is wrong

A Delivery score now reflects our mistake, not theirs.

5

Trust is lost

An employer relies on evidence that was not true.

In plain terms: the Verified Profile is only as trustworthy as the weakest message the AI Manager sends. The service exists to make every message defensible.

THE GOAL

Six Requirements Every Option Was Judged Against

30%

Hallucination control

No invented deadline, colleague, tool or file may reach a candidate.

20%

Auditability

Every statement traceable to its source, matching the verifiable evidence promise.

15%

Exact identifiers

Activity numbers, deliverable codes, file names and store codes found precisely.

15%

Complete tasks

Activity, deliverable, due point, client, voice and market practice, together.

10%

Predictable cost and speed

Tasks are generated for every candidate and every activity.

10%

Platform fit

Works with PostgreSQL, pgvector and OpenRouter, already in use.

03

The Job Simulation Market

Six established platforms reviewed from public information, what the category does well, where the gap remains, and why this service is what lets Vet2Employ fill it.

BACKGROUND STUDY

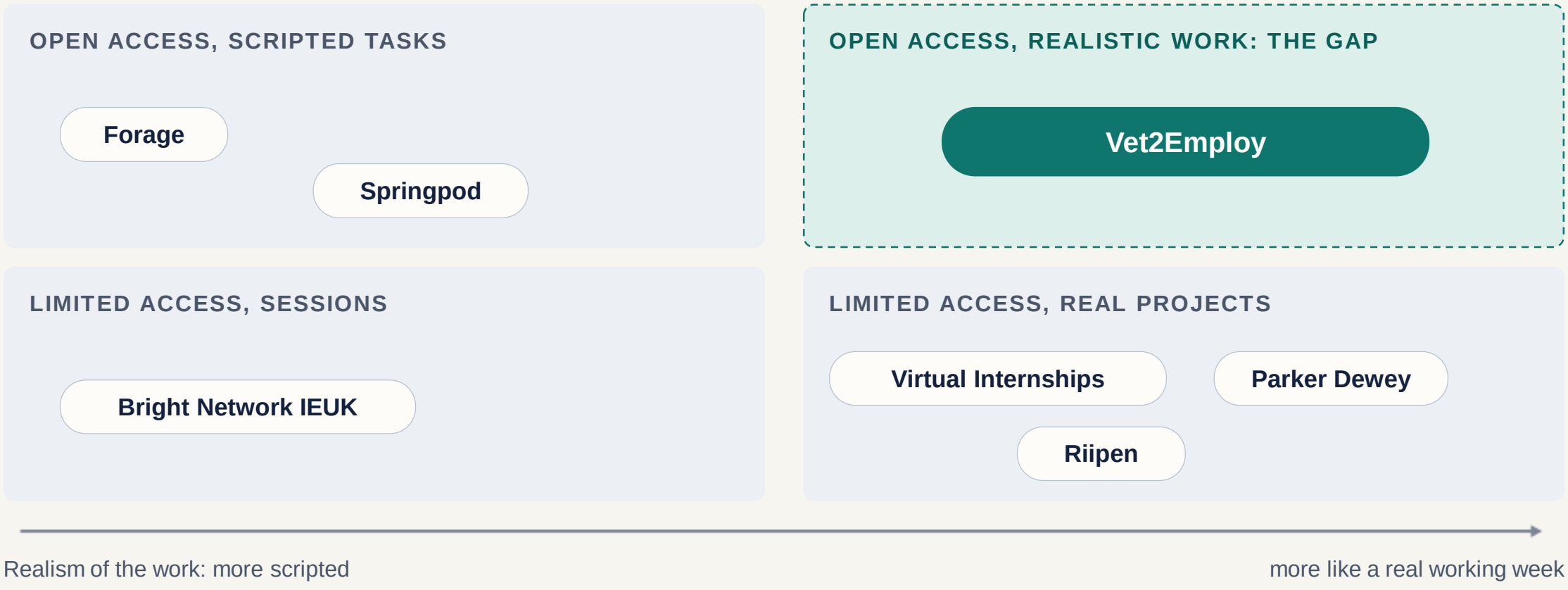
Six Platforms in the Category

Platform	Model	Commitment	Evaluated by	Leaves with
Forage	Free, on demand simulations with employers; acquired by EAB in 2024	Self paced, short	Completion only	Certificate
Virtual Internships	Remote projects with real host companies, plus coaching	1 to 4 months, 120 hours minimum	Host supervisor	Certificate and report
Riipen	Employer projects embedded in coursework	Set by the course	Educator and employer	Portfolio work
Parker Dewey	Paid micro internships posted by employers	5 to 40 hours	Posting employer	Pay and a contact
Springpod	UK virtual work experience simulations	3 to 6 hours each	Completion	Certificate
Bright Network IEUK	Scheduled virtual programme of sector sessions	Four days in June	Attendance	Certificate

Sources: Campus Technology on the EAB acquisition of Forage; Virtual Internships FAQ; Riipen; Parker Dewey via Oregon State University; Springpod via the University of Liverpool; Bright Network. Public information as of October 2026.

POSITIONING

Access Versus Realism: Where the Gap Sits



Positions are qualitative, drawn from each platform's public description

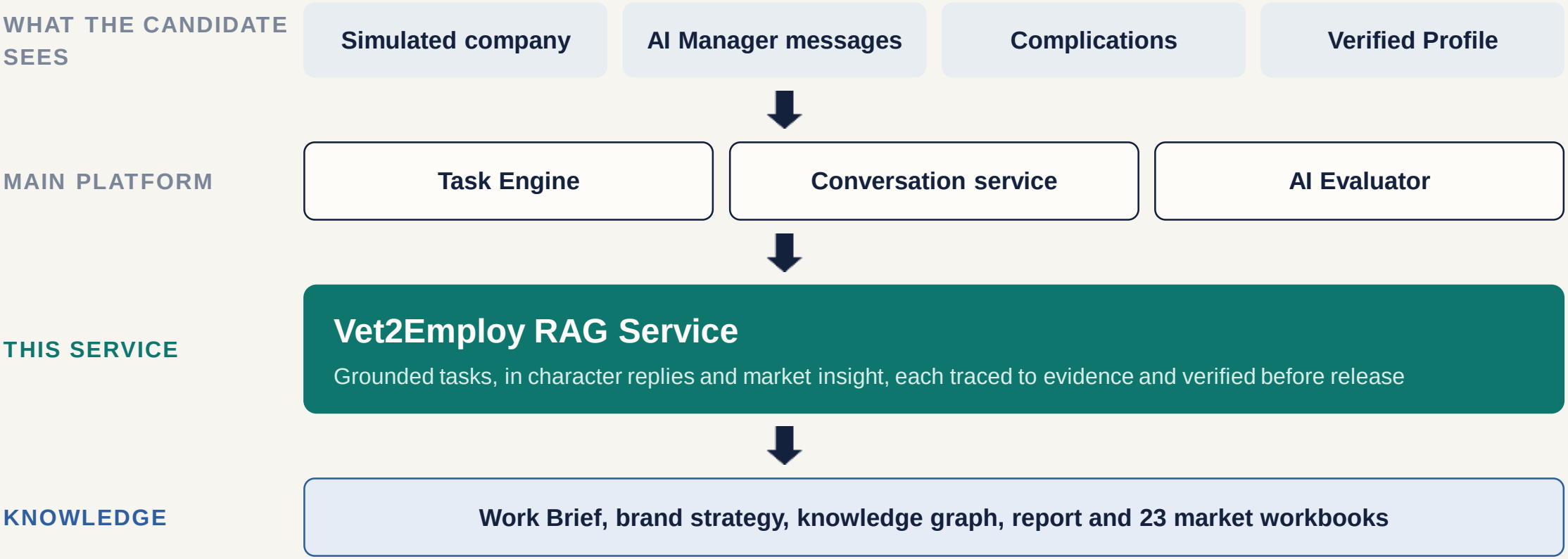
DISTINCTIVENESS

What Makes Vet2Employ Different

Dimension	Prevailing approach	Vet2Employ
Environment	A task list or a set of sessions	A simulated company with a team, a client and a weekly rhythm
Management	Static briefs or a host supervisor	An AI Manager who assigns, answers and gives feedback in character
Realism	Short scripted tasks or variable projects	A full week of activities, enriched by 521 researched UK roles
Adversity	Rarely modelled	Deliberate complications, assessed as carefully as output
Evaluation	Completion, attendance or opinion	Five weighted competencies, rubric plus AI evaluator
Evidence	A certificate	A Verified Profile with hashed deliverables and scores
Access	Limited by employers or dates	Any candidate, any time

WHERE THIS SERVICE FITS

The Realism Layer Under the Distinctive Features



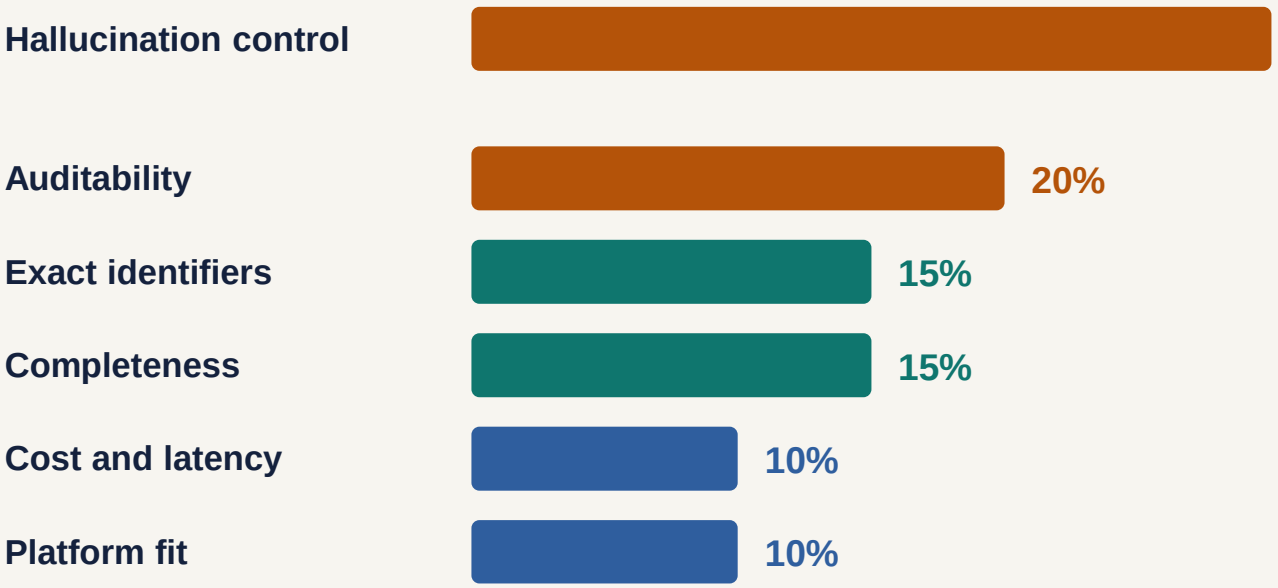
04

Five RAG Architectures, Analysed

Each candidate gets the same seven slides: the diagram, a plain explanation, the technical detail, strengths, limits, a real Vet2Employ test case and a scorecard.

EVALUATION METHOD

One Weighted Scorecard, Applied Equally



Scale of 1 to 5

1 means the requirement is not met by design; 5 means it is met by a mechanism that can be tested.

Weighted total

Sum of score times weight. A perfect design scores 5.00.

Why trust leads

Half the weight sits on truth and traceability, because the Verified Profile is the product.

THE CANDIDATES

Five Architectures at a Glance

A

Naive vector RAG

Embed the question, take the closest chunks, paste them into one prompt.

One search, one call

B

Hybrid search and rerank

Combine meaning search with keyword search, then reorder the results.

Two searches, one call

C

Agentic RAG

The model decides what to search for, in a loop, until it is satisfied.

Many searches, many calls

D

GraphRAG

Extract entities and relations into a graph, then search the graph.

Graph build, graph query

E

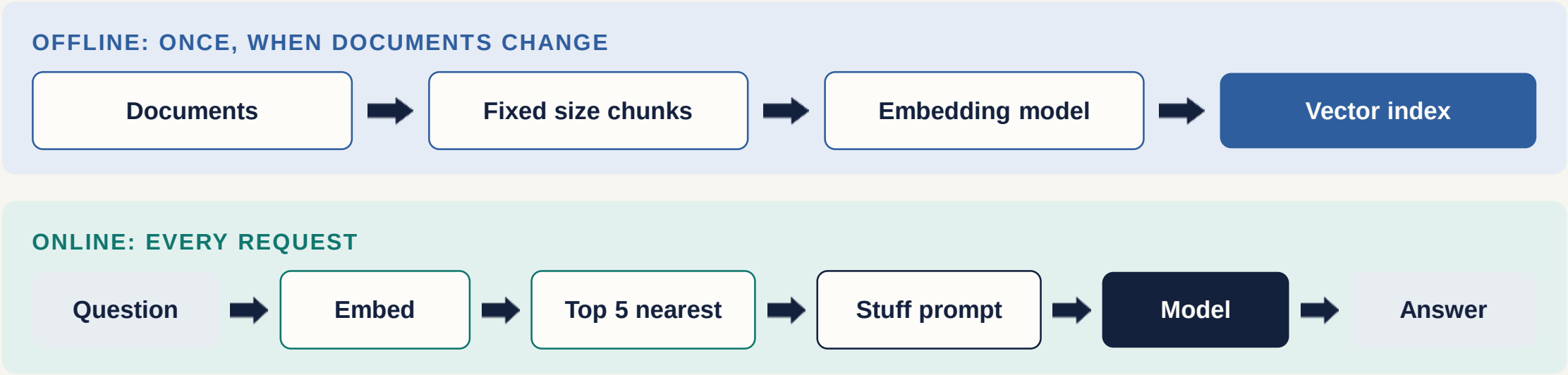
Planned and verified hybrid RAG

A fixed plan of searches per purpose, then a seven layer check of every claim.

Selected

ARCHITECTURE A | DIAGRAM

Naive Vector RAG



Note what is missing: no keyword search, no filters, no reranking, no check of the answer.

ARCHITECTURE A | IN PLAIN WORDS

A Librarian Who Judges Books by Their Feel

You ask a question. The librarian picks the five pages that *feel* most like your question, hands them to a writer, and the writer answers from those pages.

Nobody checks whether the five pages were the right ones, and nobody checks whether the writer stayed inside them.

1. Good when the question uses the same words and ideas as the source.
2. Weak when the answer hinges on an exact code such as “D3” or “Activity 7”.
3. Risky when the writer fills gaps from general knowledge.

The Whole Pipeline Fits in Six Lines

```
# naive_rag.py
def answer(question):
q = embed(question)
hits = index.nearest(q, k=5)
context = "\n\n".join(h.text for h in hits)
prompt = TEMPLATE.format(context, question)
return llm(prompt)
```

Chunking	**Fixed size, no structure**
Similarity	**Cosine on one vector**
Candidates	**Top k, usually 3 to 10**
Metadata filters	**None**
Model calls	**1**
Output check	**None**

ARCHITECTURE A | STRENGTHS

Cheap, Fast and Easy to Explain

+ Lowest cost per request

One embedding and one model call. Nothing else is billed.

+ Lowest latency

A single index lookup in milliseconds, then one generation.

+ Quick to build

Any vector database and any model; a prototype in a day.

+ Good paraphrase recall

Finds “stakeholder update” when asked about “client communication”.

ARCHITECTURE A | LIMITS

Four Ways It Fails a Simulation

Misses exact codes

“Activity 3” and “Activity 8” look almost identical to an embedding.

Crowding

1,835 job market chunks can push the 163 brief chunks out of the top five.

No completeness

A task needs six kinds of fact. One search returns whichever kind is loudest.

Silent invention

With no check, a plausible deadline or colleague reaches the candidate.

ARCHITECTURE A | VET2EMPLOY TEST CASE

Request: Data Analyst Track, Activity 3 Task

1 THE TOP FIVE

- Data Analyst posting, consulting
- Data Analyst posting, healthcare
- Methodology note on analysis
- Brief: Activity 8 text
- Data Analyst posting, IT

2 WHAT IS MISSING

- Activity 3 definition
- Deliverable code and file name
- Kestrel Lane client context
- Due point in the week
- Supervisor voice

3 LIKELY OUTCOME

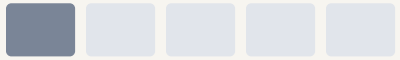
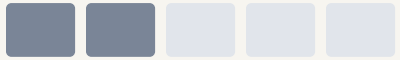
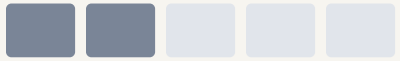
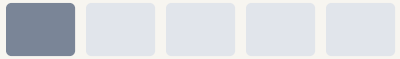
A fluent task built around the wrong activity, with a deadline and a file name the model supplied from general knowledge.

Nothing flags it.

Illustrative walk through. The crowding pattern is the one the retrieval evaluation measured before source balancing was added.

ARCHITECTURE A | SCORECARD

Naive Vector RAG Scores 2.05 out of 5

Hallucination control	30%		**0.30**
Auditability	20%		**0.40**
Exact identifiers	15%		**0.30**
Completeness	15%		**0.15**
Cost and latency	10%		**0.50**
Platform fit	10%		**0.40**

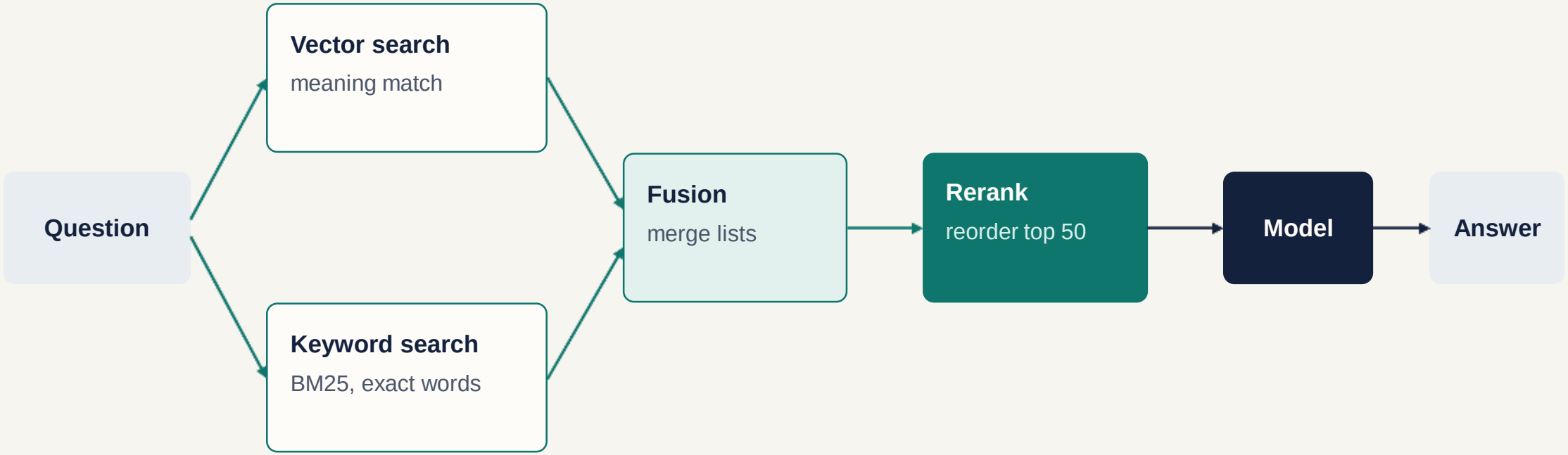
WEIGHTED TOTAL

2.05

Strong on cost, weakest exactly where the product needs strength: truth and completeness.

ARCHITECTURE B | DIAGRAM

Hybrid Search and Rerank



ARCHITECTURE B | IN PLAIN WORDS

Two Librarians and an Editor

THE FIRST LIBRARIAN

Finds pages that mean the same thing.

Understands that “client update” and “stakeholder message” are related.

THE SECOND LIBRARIAN

Finds pages that use the same words.

Never confuses “Activity 3” with “Activity 8”, or “D3” with “D2”.

THE EDITOR

Merges both piles and puts the best on top.

Better pages reach the writer. The writer is still unchecked.

ARCHITECTURE B | TECHNICAL DETAIL

Two Channels, One Fused Ranking

Dense channel

Approximate nearest neighbour search over embeddings, usually an HNSW index.

Sparse channel

BM25 or database full text search; rewards rare exact terms.

Reranker

A cross encoder reads question and passage together and rescores the shortlist.

RECIPROCAL RANK FUSION

$$\text{score}(d) = \Sigma\ 1 \div (k + \text{rank}(d))$$

Sum over each channel that returned the passage. k dampens the head of each list; 60 is the standard value and the one this service uses.

Ranks, not raw scores, are fused, so cosine and BM25 never need a common scale.

ARCHITECTURE B | STRENGTHS

The Industry Baseline for Good Reasons

+ Exact identifiers found

Keyword search catches activity numbers, file names and store codes.

+ Meaning still covered

The vector channel keeps paraphrase recall from Architecture A.

+ Fits PostgreSQL natively

pgvector plus built in full text search; no second database.

+ Still predictable

Two queries in parallel and one model call; cost stays flat.

ARCHITECTURE B | LIMITS

Better Retrieval Is Not the Same as a True Answer

Still one question

One query cannot ask for activity, client, schedule and voice at once.

No source authority

A job posting can outrank the Work Brief that defines the programme.

Generation unchecked

Good context reduces invention, it does not prevent it.

No citations by default

The answer cannot be traced back to the passages it used.

ARCHITECTURE B | VET2EMPLOY TEST CASE

Same Request: Activity 3, Data Analyst Track

1 NOW RETRIEVED

- Brief: Activity 3 definition
- Brief: deliverable for Activity 3
- Data Analyst posting, consulting
- Data Analyst posting, IT
- Methodology note

2 STILL MISSING

- Kestrel Lane client context
- Weekly schedule and due point
- Supervisor voice
- Rubric and planted issues

3 LIKELY OUTCOME

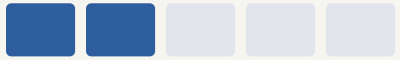
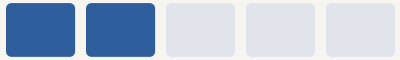
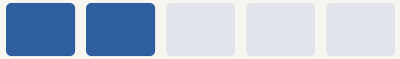
The right activity, the right deliverable. The deadline, the client detail and the manager’s tone are still guessed.

Better, still unverified.

Illustrative walk through. Keyword search is what lifted the Activity 3 passage from semantic rank 13 to the top in the real evaluation.

ARCHITECTURE B | SCORECARD

Hybrid Search and Rerank Scores 2.70 out of 5

Hallucination control	30%		**0.60**
Auditability	20%		**0.40**
Exact identifiers	15%		**0.60**
Completeness	15%		**0.30**
Cost and latency	10%		**0.40**
Platform fit	10%		**0.40**

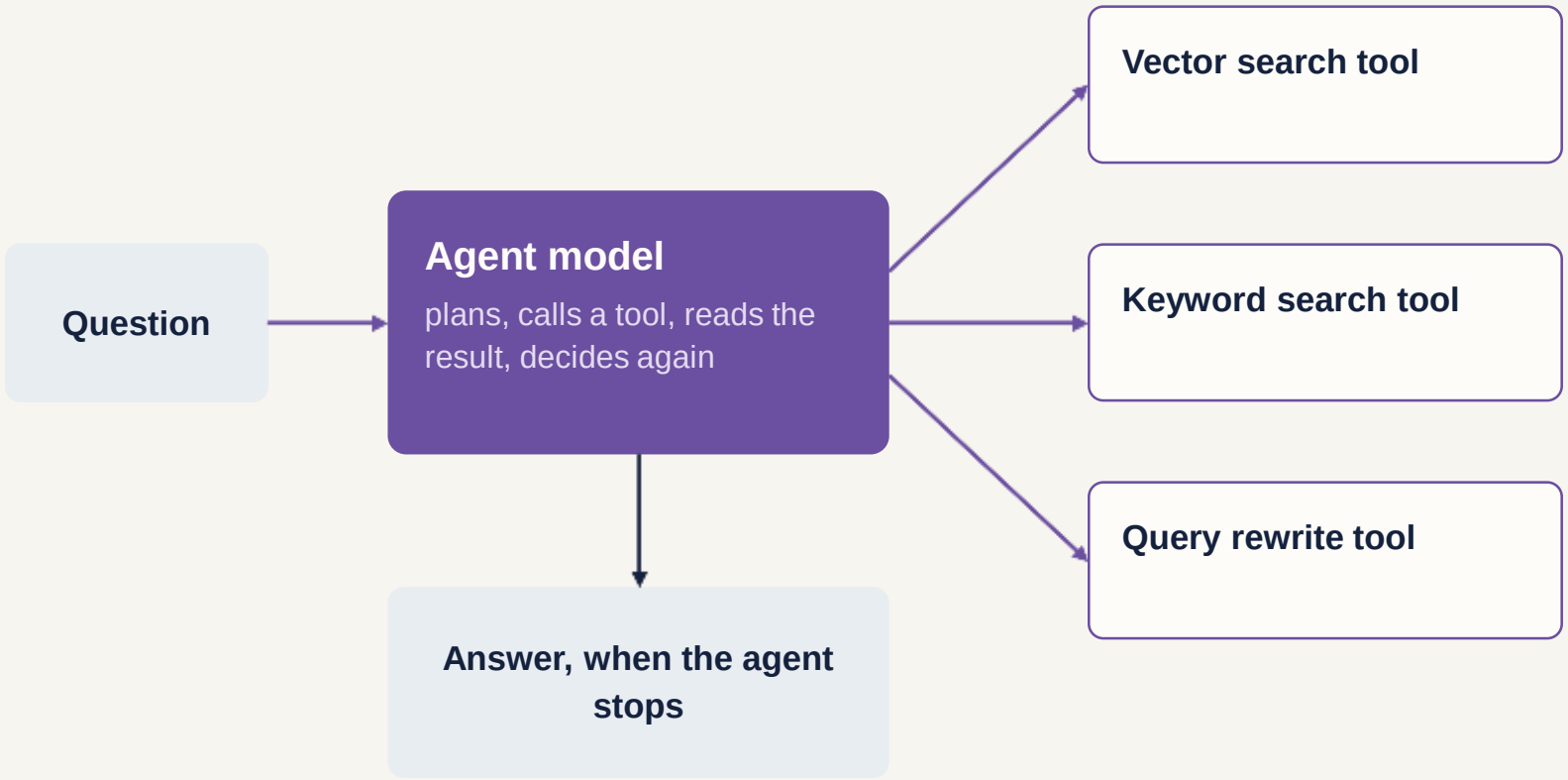
WEIGHTED TOTAL

2.70

A real gain on identifiers. Its retrieval core survives into the chosen design.

ARCHITECTURE C | DIAGRAM

Agentic RAG



THE LOOP

Each turn is a model call. The agent chooses the tool, the query and when to stop. Typical runs take 3 to 10 turns.

Results flow back into the agent’s context every turn.

ARCHITECTURE C | IN PLAIN WORDS

A Researcher Who Decides What to Look Up

Instead of one search, the writer becomes a researcher: look something up, read it, decide what is still missing, look again, and stop when it feels complete.

Powerful for open questions. But “feels complete” is the model’s judgement, and every extra look is another paid call.

Turn 1 “Search: Activity 3 Data Analyst”

Turn 2 “Search: Kestrel Lane client”

Turn 3 “Search: Data Analyst SQL tools”

Turn 4 “I have enough.” (schedule never fetched)

ARCHITECTURE C | TECHNICAL DETAIL

A Tool Calling Loop With a Turn Budget

```
# agentic_rag.py
messages = [system, user(question)]
for turn in range(MAX_TURNS):
reply = llm(messages, tools=TOOLS)
if not reply.tool_calls:
return reply.text
for call in reply.tool_calls:
result = run_tool(call)
messages.append(result)
```

Model calls	**3 to 10 per request**
Context growth	**Grows every turn**
Search plan	**Decided by the model**
Stop rule	**Model judgement or cap**
Repeatability	**Low: paths differ per run**
Output check	**Optional self critique**

ARCHITECTURE C | STRENGTHS

Flexible Enough for Questions Nobody Planned

+ Multi step reasoning

Can chain lookups: find the client, then the client’s systems.

+ Better completeness

Several searches cover more ground than one.

+ Query repair

Rewrites a search that came back empty.

+ Open ended use

Suits free form candidate questions with no fixed shape.

ARCHITECTURE C | LIMITS

Unpredictable Where We Need Predictable

Cost multiplies

3 to 10 model calls per task, with a context that grows every turn.

Latency multiplies

Sequential turns mean many seconds before the candidate sees a task.

Different every run

Two candidates on the same activity may get differently researched tasks.

Self grading

The model that wrote the answer judges whether it is complete.

ARCHITECTURE C | VET2EMPLOY TEST CASE

Same Request, Run Twice

RUN 1 | CANDIDATE A

Searches: activity, client, tools, rubric

Model calls: 6

Has rubric detail; no schedule

Due point invented

RUN 2 | CANDIDATE B

Searches: activity, schedule, voice

Model calls: 4

Has schedule; no rubric or client

Client detail invented

THE FAIRNESS PROBLEM

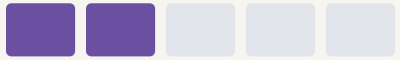
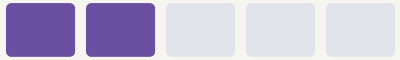
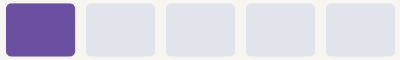
Two candidates, same activity, different quality of task.

A scored programme cannot accept that.

Illustrative walk through of the behaviour described on the previous slides.

ARCHITECTURE C | SCORECARD

Agentic RAG Scores 2.45 out of 5

Hallucination control	30%		**0.60**
Auditability	20%		**0.40**
Exact identifiers	15%		**0.45**
Completeness	15%		**0.60**
Cost and latency	10%		**0.10**
Platform fit	10%		**0.30**

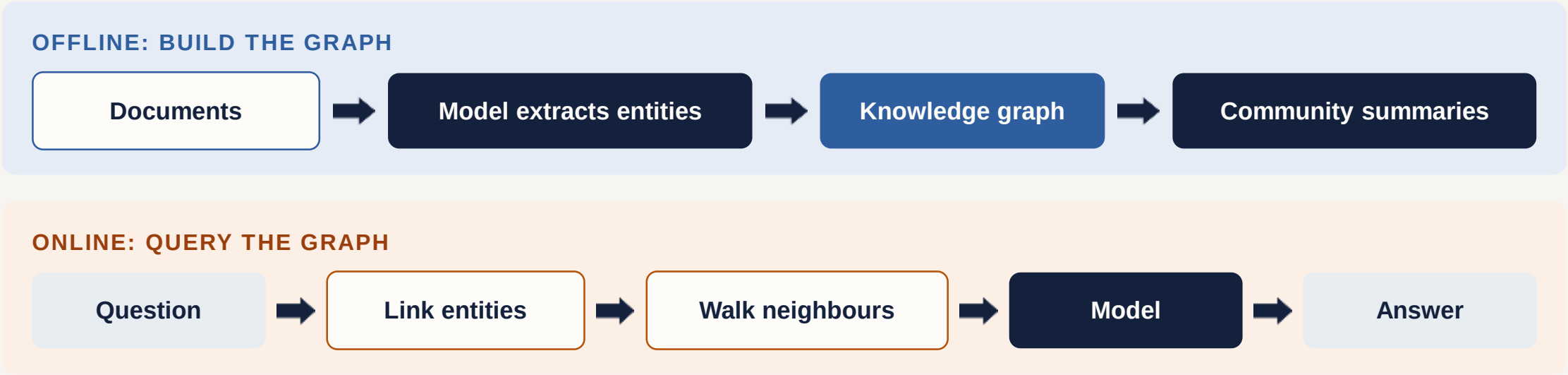
WEIGHTED TOTAL

2.45

Its best idea, several targeted searches, is kept, but made deterministic.

ARCHITECTURE D | DIAGRAM

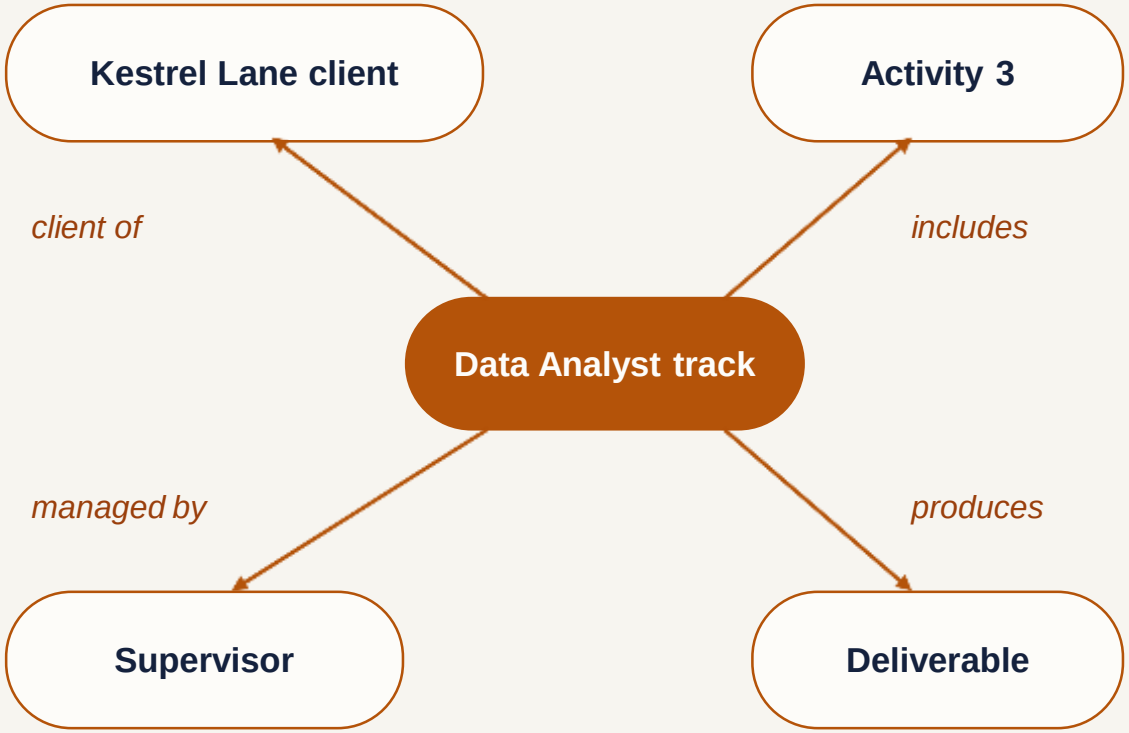
GraphRAG



Note: the model is used twice; once offline to build the graph, once online to answer.

ARCHITECTURE D | IN PLAIN WORDS

A Map of Who and What Connects to What



Before answering, a model reads every document and draws a map of the people, companies and things in it, and how they relate.

To answer, it starts at the things you named and walks the map. Excellent for “how is X connected to Y”. The map itself is drawn by a model, so it can contain mistakes.

ARCHITECTURE D | TECHNICAL DETAIL

Extraction, Communities and Two Query Modes

1

Extraction

A model reads each chunk and emits entities and relations as triples. 3,194 chunks means 3,194 extra model calls on every rebuild.

2

Communities

A clustering algorithm such as Leiden groups related nodes; a model writes a summary of each community.

3

Query modes

Local: start at linked entities, walk neighbours. **Global:** map and reduce over community summaries for broad questions.

Infrastructure: a graph store (or graph tables) alongside the vector store, plus a rebuild job whenever documents change.

ARCHITECTURE D | STRENGTHS

The Strongest Rival on Paper

+ Relationship questions

Answers “which supervisor owns which client” directly from edges.

+ Traceable structure

Nodes and edges are inspectable, which helps auditability.

+ Corpus wide themes

Global mode can summarise culture across all 521 roles.

+ Natural fit with the graph PDF

Vet2Employ already has a knowledge graph document to seed it.

ARCHITECTURE D | LIMITS

A Model Built Map Can Be Wrong Before Anyone Asks

Extraction errors persist

A wrong edge becomes a “fact” for every future request.

Heavy rebuilds

Every workbook update means re extraction and re clustering.

Mostly tabular data

The 23 workbooks are already structured; extracting a graph adds little.

Still no output check

A better context, but the generated task is not verified.

ARCHITECTURE D | VET2EMPLOY TEST CASE

Same Request, Through the Graph

1 WHAT WORKS

Links track to client, Kestrel Lane

Links client to supervisor

Links activity to deliverable

2 WHAT CAN GO WRONG

Extraction merges “Activity 3” across tracks into one node

Market detail is summarised away by communities

The exact file name lives in a table cell, not an edge

3 LIKELY OUTCOME

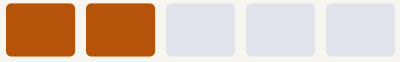
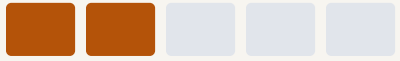
Strong structure and names. Specific numbers and file names less reliable. Still no gate before release.

Good context, unverified output.

Illustrative walk through. Merged nodes across tracks are a known GraphRAG failure when entity names repeat.

ARCHITECTURE D | SCORECARD

GraphRAG Scores 3.00 out of 5

Hallucination control	30%		**0.90**
Auditability	20%		**0.80**
Exact identifiers	15%		**0.45**
Completeness	15%		**0.45**
Cost and latency	10%		**0.20**
Platform fit	10%		**0.20**

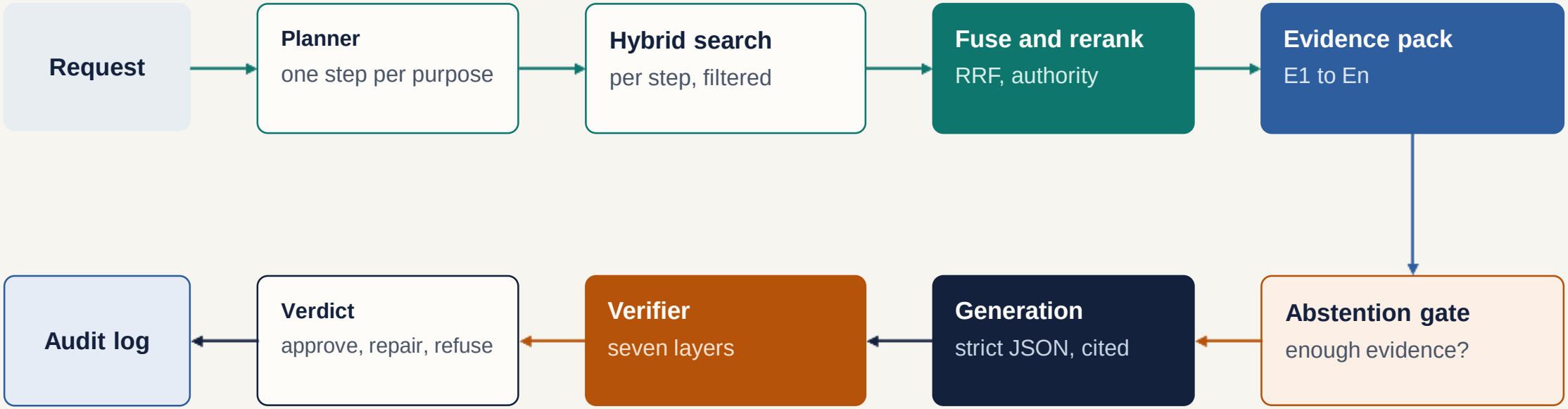
WEIGHTED TOTAL

3.00

The runner up. Kept on the roadmap as a later extension, not a starting point.

ARCHITECTURE E | DIAGRAM

Planned and Verified Hybrid RAG



■ retrieval ■ data ■ model ■ verification and safety

ARCHITECTURE E | IN PLAIN WORDS

A Newsroom: Checklist, Researchers, Writer and Fact Checker

1

The checklist

Every task story needs the same facts: activity, deliverable, client, schedule, voice, rubric, market practice.

2

The researchers

One researcher per item, each searching by meaning and by exact words, preferring the most official source.

3

The writer

Writes only from the numbered notes, and footnotes every sentence. No notes, no story.

4

The fact checker

A different person checks every footnote. Anything unsupported is fixed, cut or the story is spiked.

ARCHITECTURE E | TECHNICAL DETAIL

Deterministic Retrieval, Independent Verification

Stage	Mechanism	Setting
Plan	Purpose specific steps with filters by track, activity, source, role and sector	Up to 13 step purposes
Search	pgvector HNSW cosine plus PostgreSQL full text, per step	40 candidates per channel
Fuse	Reciprocal rank fusion	k = 60
Rerank	Fused score, source authority and query coverage	0.65, 0.15, 0.20
Diversify	Exact and near duplicate removal, employer cap	Jaccard 0.85, 2 per employer
Pack	Numbered evidence E1 to En within budget	7,000 tokens, 24 items
Generate	JSON schema output; every item cites evidence ids	Claude Sonnet 4.5
Verify	Deterministic checks plus a separate entailment judge	Claude Haiku 4.5

ARCHITECTURE E | STRENGTHS

Each Rival’s Best Idea, Without Its Weakness

+ From B: hybrid search

Exact codes and paraphrases are both found, inside one PostgreSQL.

+ From C: many searches

Several targeted searches, but planned in code, not chosen by a model.

+ From D: structure

Metadata filters on track, activity and role give graph like precision.

+ Unique: verification

The only option that checks every claim before a candidate sees it.

ARCHITECTURE E | LIMITS

The Honest Costs of the Chosen Design

Two or three model calls

Generation plus a judge, plus one repair when needed.

Plans need upkeep

A new kind of request needs a new retrieval plan in code.

Refusals are visible

The platform must handle 422 and 503 answers gracefully.

More moving parts

More code to own, offset by 96 automated tests.

ARCHITECTURE E | VET2EMPLOY TEST CASE

Same Request, Measured on the Real Knowledge Base

1 EVERY PURPOSE FILLED

Activity 3 definition and deliverable

Kestrel Lane client context

Schedule and supervisor voice

Rubric and planted issues

Market tools and culture

2 THE KEY PASSAGE

Activity 3 brief passage was only **13th** by meaning, but **1st** by keywords.

After fusion and rerank it finished on top, final score **0.937**.

3 OUTCOME

Approved

Groundedness 1.0. Every check passed. Every sentence traceable to E numbers.

Measured: offline retrieval run on the ingested knowledge base, and an end to end run with a mock OpenRouter.

ARCHITECTURE E | SCORECARD

Planned and Verified Scores 4.65 out of 5

Hallucination control	30%		**1.50**
Auditability	20%		**1.00**
Exact identifiers	15%		**0.60**
Completeness	15%		**0.75**
Cost and latency	10%		**0.30**
Platform fit	10%		**0.50**

WEIGHTED TOTAL

4.65

Full marks on both trust requirements. It pays for that with moderate cost.

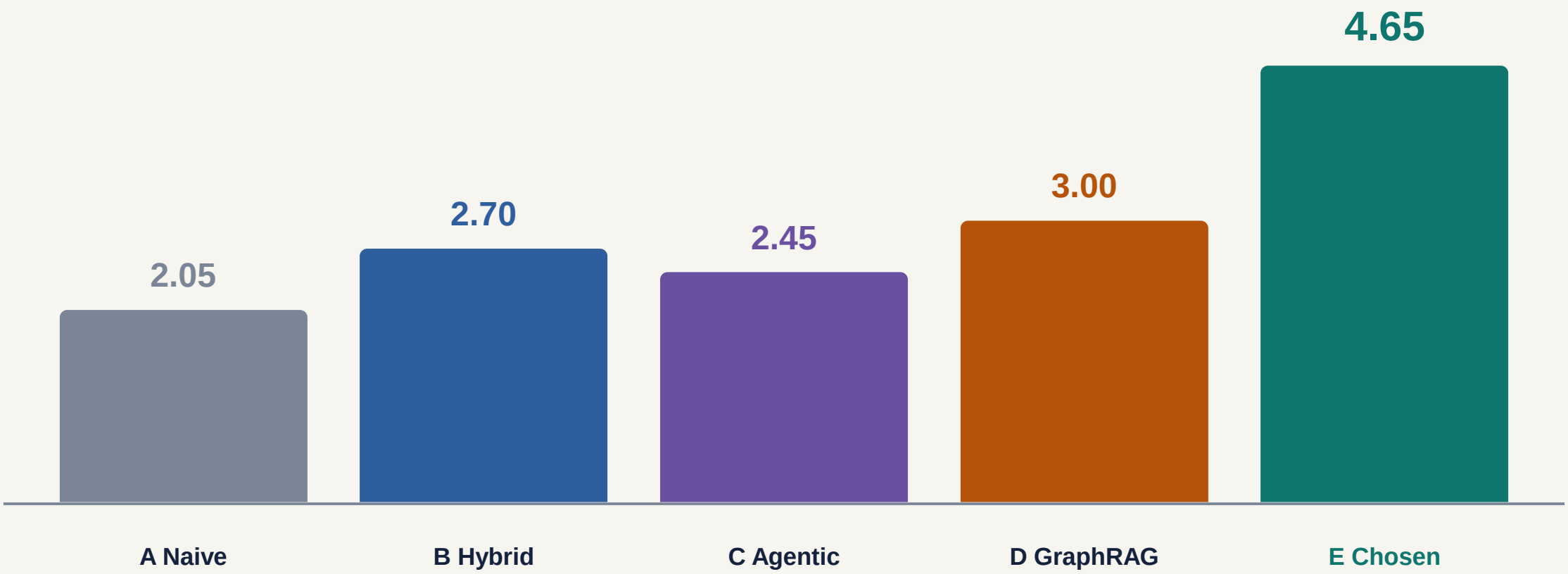
COMPARISON

The Full Weighted Scorecard

Requirement	Weight	A Naive	B Hybrid	C Agentic	D Graph	E Chosen
Hallucination control	30%	1	2	2	3	5
Auditability	20%	2	2	2	4	5
Exact identifiers	15%	2	4	3	3	4
Completeness	15%	1	2	4	3	5
Cost and latency	10%	5	4	1	2	3
Platform fit	10%	4	4	3	2	5
Weighted total	100%	2.05	2.70	2.45	3.00	4.65

COMPARISON

A Clear Winner, by 1.65 Points Over the Runner Up



Bar height is proportional to the weighted total out of 5 | Comparison and decision | 61

COMPARISON

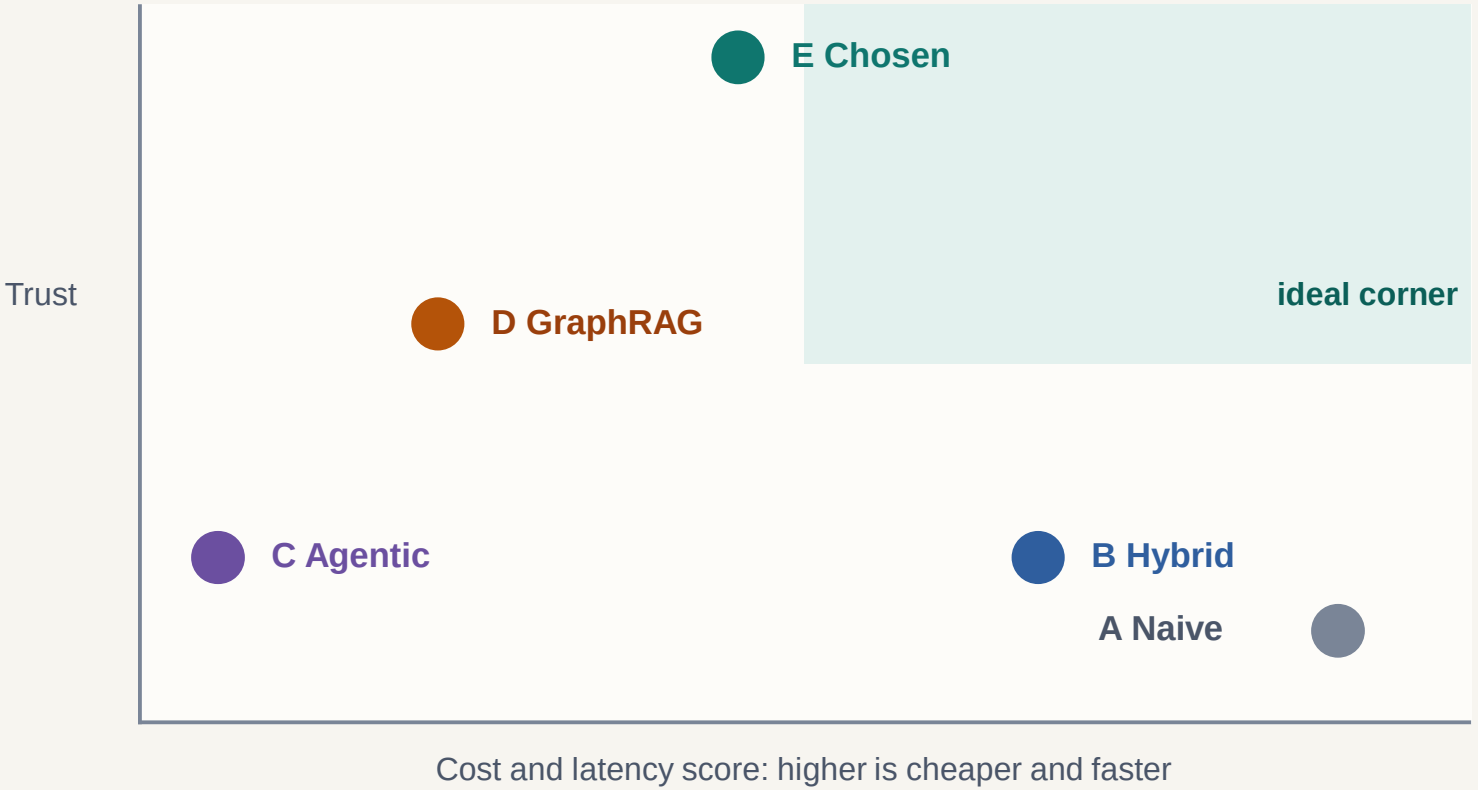
Heat Map: Where Each Design Is Strong or Weak

	A Naive	B Hybrid	C Agentic	D Graph	E Chosen
Hallucination control	1	2	2	3	5
Auditability	2	2	2	4	5
Exact identifiers	2	4	3	3	4
Completeness	1	2	4	3	5
Cost and latency	5	4	1	2	3
Platform fit	4	4	3	2	5

Darker teal is stronger, amber is weaker

TRADE OFF

Trust Versus Running Cost



Trust axis

Weighted mean of hallucination control and auditability.

Reading it

Only E reaches top trust. It sits mid cost, nearer the ideal corner than any rival.

THE DECISION

Five Reasons Architecture E Was Selected

1. **The product is verified evidence.** Only E checks every claim before release.
2. **Tasks have a known shape.** A fixed plan retrieves every needed fact, every time.
3. **Fairness between candidates.** The same activity gets the same quality of evidence.
4. **One database.** PostgreSQL with pgvector does vectors, keywords and audit.
5. **Bounded cost.** Two or three model calls, never an open ended loop.

CONSCIOUS TRADE OFFS

What We Gave Up, and Why It Is Acceptable

Given up	Cost to us	Why acceptable
Lowest possible cost	A judge call per task, about one extra small model call	The judge runs on a smaller, cheaper model
Lowest latency	Verification adds seconds	Tasks are prepared ahead of the candidate’s day
Open ended questions	Only planned request types are served	Simulation requests have known shapes
Always answering	Some requests return 422 or 503	A refusal is safer than an invented fact
Graph reasoning	No multi hop relationship queries yet	Metadata filters cover today’s needs

FUTURE PROOFING

A Path to GraphRAG Without a Rewrite

TODAY

Planned and verified hybrid

Metadata filters act as a lightweight graph: track, activity, role, sector.

STEP 1

Curated entity tables

Clients, supervisors and deliverables as rows, written by people, not extracted.

STEP 2

A graph retriever

Added as one more step type in the planner, alongside the existing ones.

UNCHANGED

Verification

Graph evidence becomes E numbers and is checked like any other.

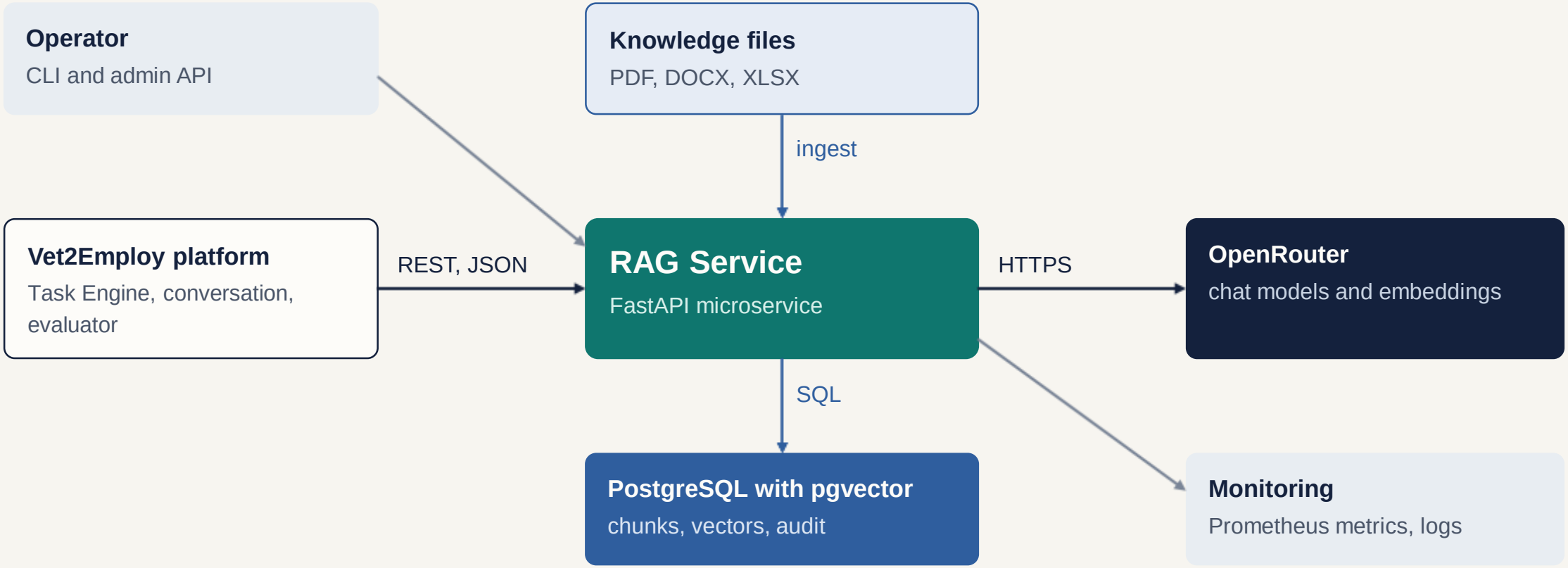
06

The Selected Architecture in Depth

System context, containers, the ten step request lifecycle, sequence diagrams, the data model, worked examples of fusion and reranking, and the seven layers of verification.

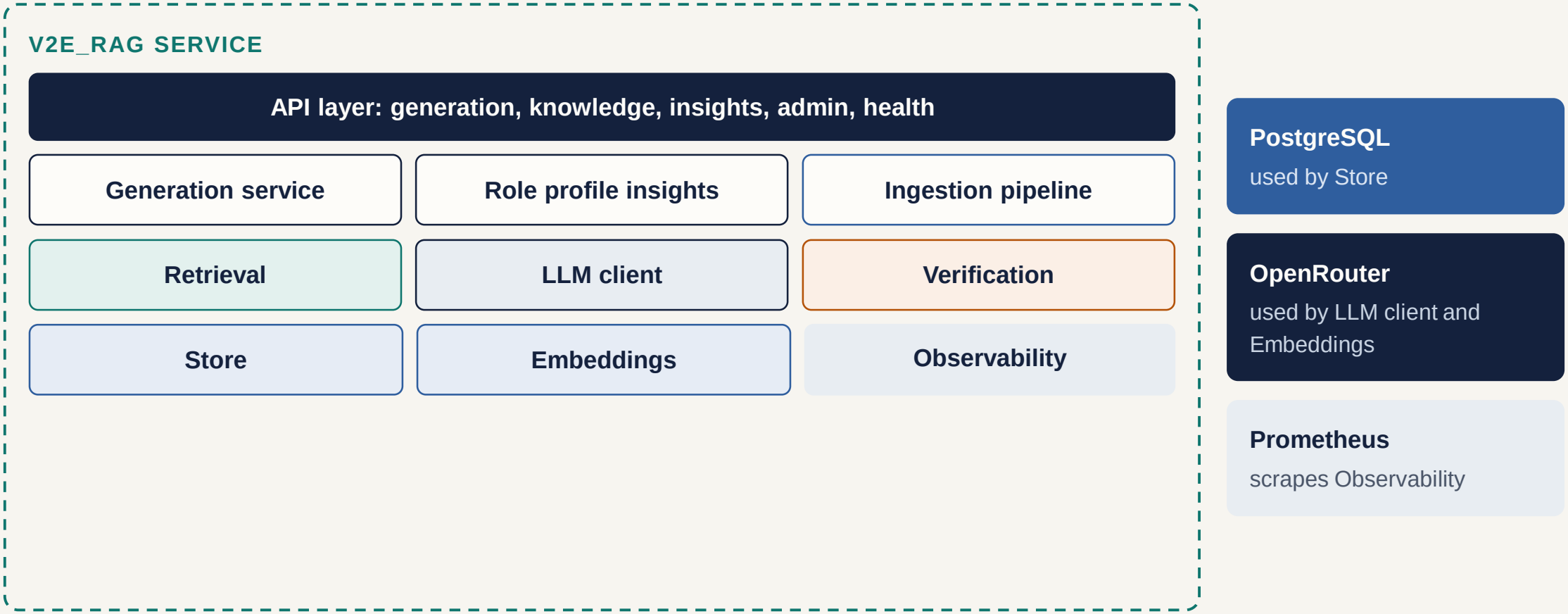
SYSTEM CONTEXT

Who Talks to the Service, and Through What



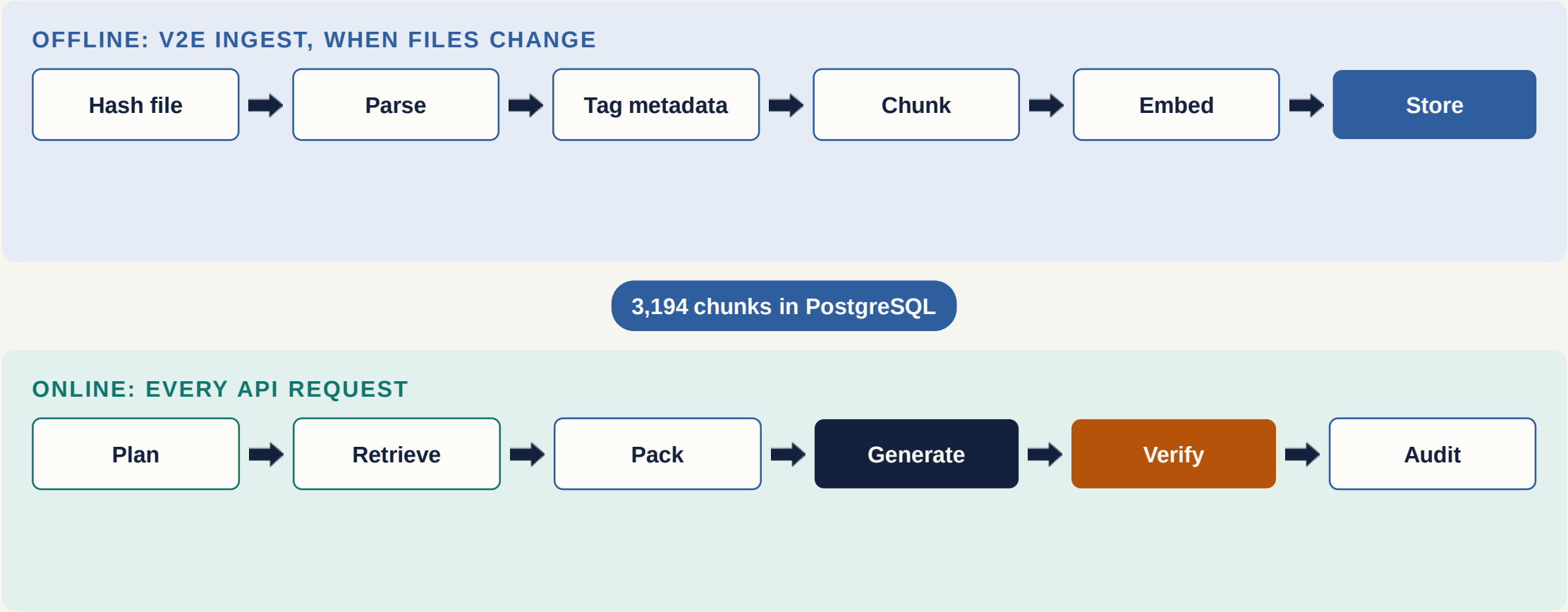
CONTAINER VIEW

Inside the Service Boundary



TWO PIPELINES

Offline Ingestion Feeds Online Generation



REQUEST LIFECYCLE

Ten Steps From Request to Verified Task

01
Validate
Strict request model

02
Plan
One step per purpose

03
Embed queries
Each unique query once

04
Search
Two channels, relaxation

05
Fuse and rerank
RRF, authority, diversity

06
Pack evidence
E1 to En, in budget

07
Abstain or continue
422 if evidence missing

08
Generate
Strict JSON with citations

09
Verify and repair
Seven layers, prune

10
Finalise and audit
Brand rules, record

STEP 01 OF 10 | VALIDATE

Reject a Bad Request Before It Costs Anything

The platform says which track, which activity and what kind of task. The service checks it makes sense first.

- Unknown fields are refused
- Track and role family must agree
- An activity number needs a track
- Manager name is supplied, never invented

```
# generation/requests.py
class TaskRequest(_Strict):
track: TrackCode | None
role_family: RoleFamily | None
task_type = CORE_ACTIVITY
activity_number: int # 1 to 30
manager_name: str | None
include_market_evidence = True
# _Strict: extra="forbid"
# invalid input returns HTTP 422
```

STEP 02 OF 10 | PLAN

The Planner Writes a Shopping List of Facts

PLAN_TASK FOR DA, ACTIVITY 3, CORE ACTIVITY

activity_definition	required
deliverables	required
client_context	required
schedule · supervisor_voice · rubric	optional
programme_norms	optional
market_responsibilities · market_tools	optional
market_collaboration · culture_signals	optional
brand_voice	optional

Each step carries

A name, a purpose, a query, filters, a limit, a required flag and a relaxation ladder.

Complication tasks

Add a required planted_issues step, so complications come from the brief.

retrieval/planner.py

STEP 03 OF 10 | EMBED QUERIES

Turn Each Distinct Question Into a Vector, Once

12 STEP QUERIES, ACTIVITY 3

- market_responsibilities: “data analyst”
- market_tools: “data analyst”
- market_collaboration: “data analyst”
- culture_signals: “data analyst”
- plus the brief and brand queries

DEDUPLICATE

Identical queries share one embedding

then all are embedded in parallel

```
# evidence.py
unique = dict.fromkeys(
step.query for step
in plan.steps)
vectors = await gather(
embed_query(q) ...)
```

Model: text embedding 3 small via OpenRouter, 1,536 dimensions, with a content hash cache for document embeddings.

STEP 04 OF 10 | SEARCH

Every Step Runs Two Searches at Once

SEMANTIC CHANNEL

pgvector HNSW, cosine distance

Top 40 by meaning, inside the step’s filters.

LEXICAL CHANNEL

PostgreSQL full text, ts_rank

Top 40 by exact words, inside the same filters.

FILTERS AVAILABLE TO EVERY STEP

source type | track | facet | activity number | role family | job level | sector | untracked only

Too few hits?

Market steps climb a relaxation ladder: drop sector, then level, until the minimum is met.

STEP 05 OF 10 | FUSE AND RERANK

Relevance, Authority and Coverage, Then Diversity

final = 0.65 × fused ÷ max + 0.15 × (6 − authority) ÷ 5 + 0.20 × coverage

Fused relevance

RRF score from both channels, scaled so the best hit is 1.0.

Authority

Level 1 programme brief scores 1.0; level 5 engineering notes score 0.2.

Coverage

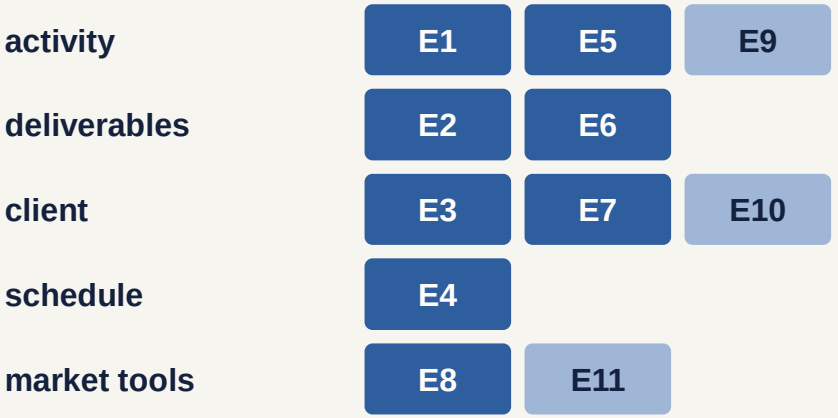
Share of the query’s words that appear in the passage.

Then diversity

No duplicates, no near duplicates, at most 2 per employer.

STEP 06 OF 10 | PACK EVIDENCE

Round Robin, So Every Purpose Gets a Seat



Illustrative order. Required steps are admitted first, then one item per step per round, until 7,000 tokens or 24 items.

Why round robin

One chatty purpose cannot fill the budget and starve the others.

Deduplicated twice

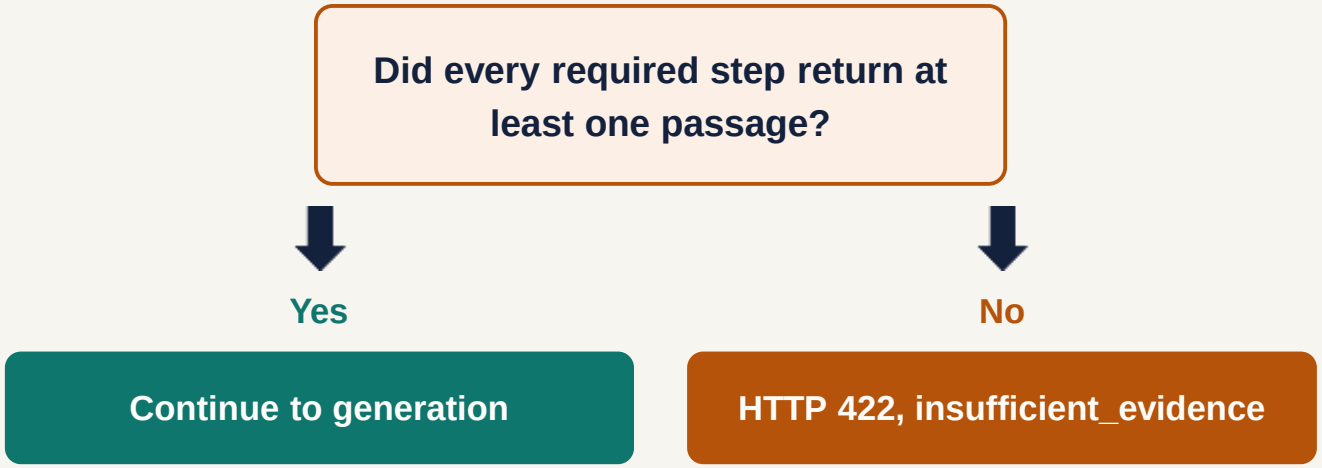
By chunk id and by content hash, across all steps.

Stable ids

E numbers are what the model must cite and the verifier will check.

STEP 07 OF 10 | ABSTAIN OR CONTINUE

No Evidence, No Model Call




```
# response body
{ "error": "insufficient_evidence",
"details": {
"missing": ["client_context"],
"diagnostics": { ... } } }
```

The platform learns exactly which purpose had no evidence. The attempt is still written to the audit table.

STEP 08 OF 10 | GENERATE

The Model Fills a Form, Not a Blank Page

WHAT THE MODEL RECEIVES

System prompt: 8 grounding rules, voice rules, task type guidance

The request fields

EVIDENCE: E1 to En with source and citation

JSON schema, strict mode, temperature 0.2

```
# what comes back
{ "title": "...",
"objective": { "text": "...",
"evidence_ids": ["E1"] },
"steps": [ { "text": "...",
"evidence_ids": ["E1","E5"] } ],
"deliverables": [ ... ],
"open_questions": [ ... ],
"message": "..." }
```

If the JSON fails schema validation, the model gets one retry at temperature 0 with the exact errors. A second failure returns HTTP 502.

STEP 09 OF 10 | VERIFY AND REPAIR

Check, Repair Once, Then Prune or Refuse

VERIFY

Split into claims. Run citation, number, entity, tool, file name and style checks, then the entailment judge.

→

REPAIR

Any major problem? Send the problems back to the model once, with the same evidence.

→

PRUNE

Still unsupported? Remove those items, if the task stays coherent.

→

REFUSE

More than 35% pruned, or core items lost? HTTP 422, grounding_failed.

approved: nothing unsupported

approved_with_pruning: small cuts

rejected: refused with reasons

STEP 10 OF 10 | FINALISE AND AUDIT

Polish the Words, Record Everything

FINALISE_TASK

~~chatbot~~	➡	**AI Manager**
~~certificate~~	➡	**Verified Profile**
a dash	➡	**a comma**

GENERATION_AUDIT ROW

audit id and timestamp

operation and final status

full request

evidence ids, chunk ids, citations

response and verification report

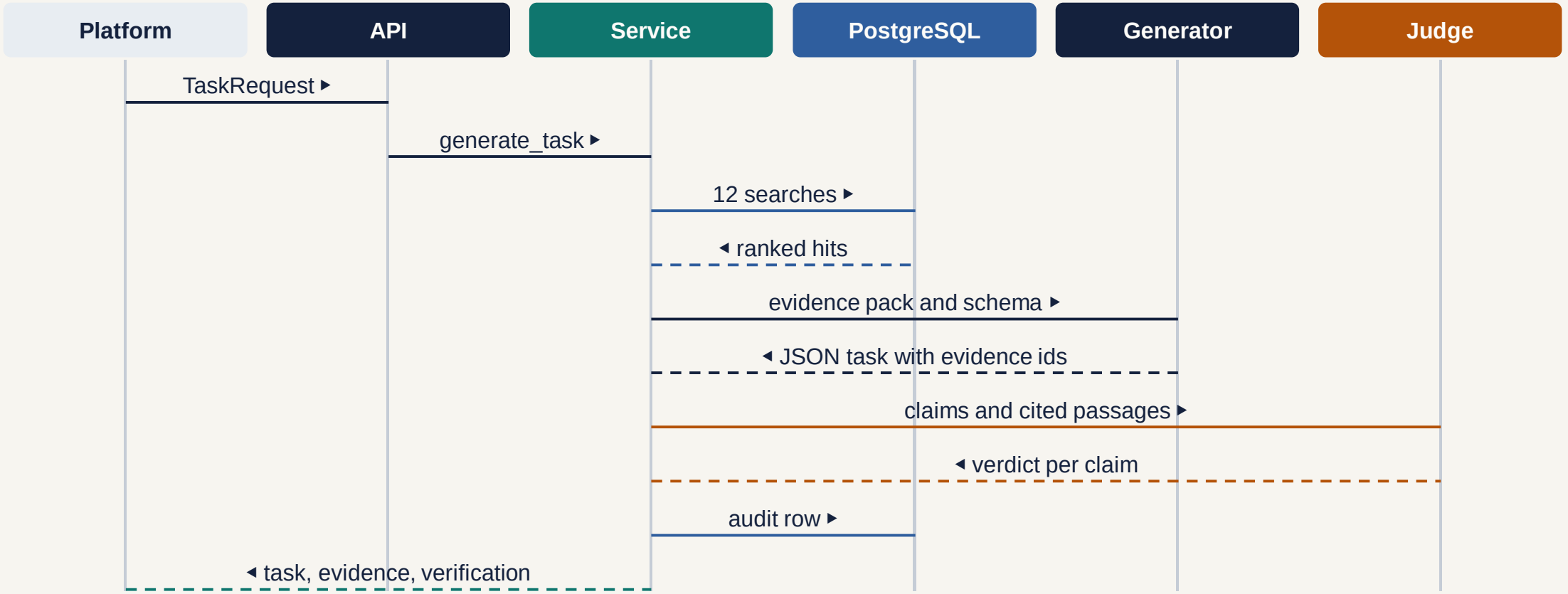
model, latency, token usage

prompt version 2026.10.1

An audit failure is logged but never breaks the candidate’s request.

SEQUENCE | POST /V1/TASKS/GENERATE

One Task, Message by Message



FLOW | POST /V1/MANAGER/REPLY

How the AI Manager Answers a Candidate

Candidate message → Redact personal data → Plan and retrieve → Evidence that answers it?

NO: ABSTAIN IN CHARACTER

“Thanks. I want to give you an accurate answer rather than a quick one, so I will check and come back to you.”

needs_human_review is set to true. No model is called.

YES: GENERATE AND VERIFY

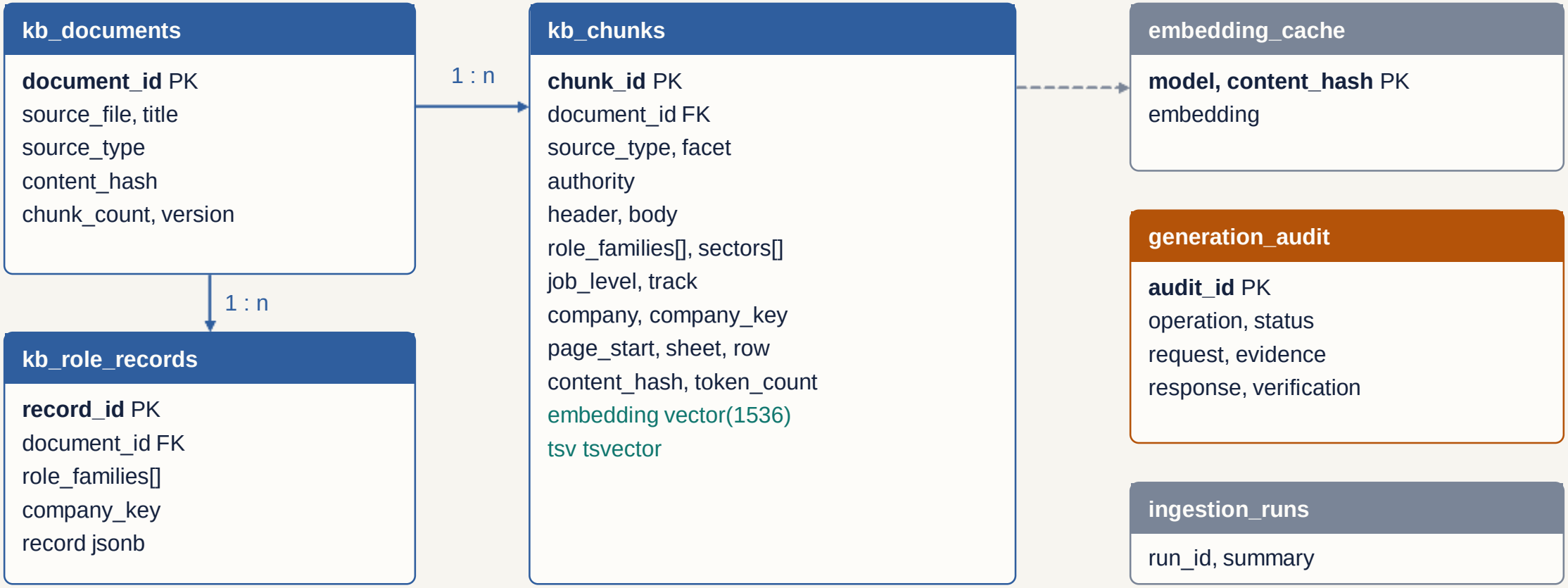
Reply under 180 words, in the supervisor’s voice, with every claim cited.

Verified like a task. If pruning would break the reply, it falls back to the in character abstention.

Wellbeing, fairness or safeguarding topics are flagged for human review.

DATA MODEL

Six Tables in One PostgreSQL Database



EVIDENCE PACK ANATOMY

What One Evidence Item Looks Like to the Model

```
[E1] (programme_brief |
track_activity |
document title, section, page)
Activity 3 ... Objective ...
You are done when ...
[E2] (programme_brief |
track_deliverables | ...)
...
```

[E1] Stable id the model must cite and the verifier checks.

source Tells the model which grounding rule applies.

facet What kind of fact the passage holds.

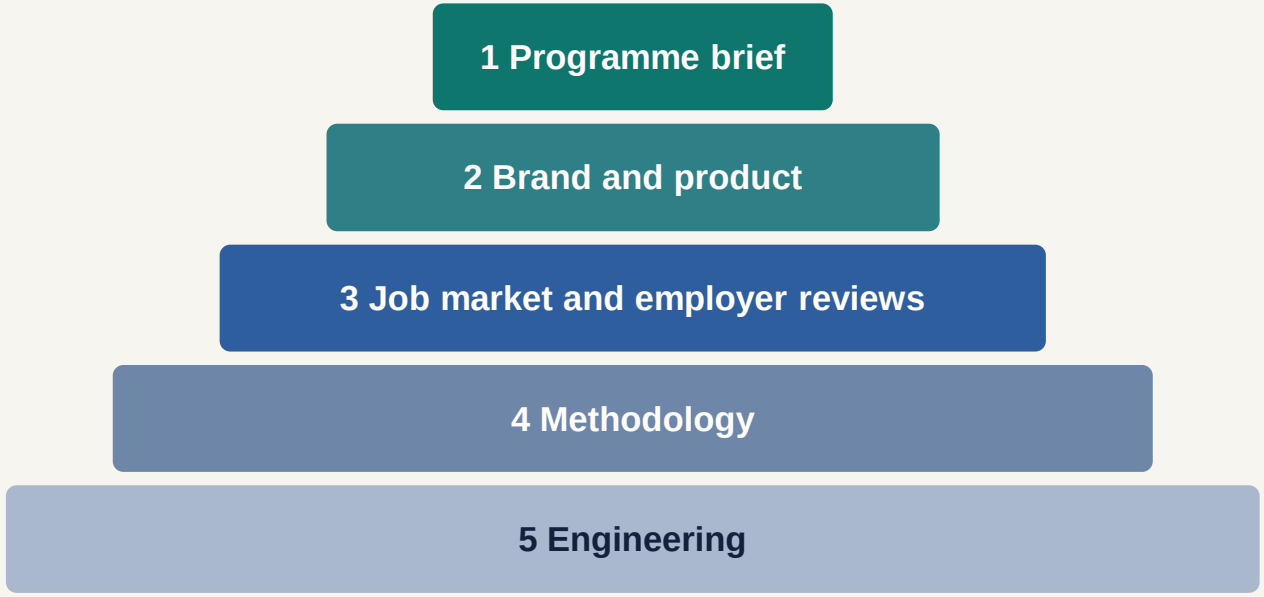
citation Human readable label, also returned to the platform.

text The chunk itself, header plus body.

Format produced by EvidencePack.render in domain/models.py. Job postings cite employer, job title and workbook instead of pages.

SOURCE AUTHORITY

When Sources Disagree, the Higher Level Wins



In the reranker

(6 − level) ÷ 5: level 1 scores 1.0, level 5 scores 0.2, worth 15% of the final score.

In the prompt

Rule 5: the brief always overrides any other source. Rule 6: market data is for realism only.

RETRIEVAL PLAN FOR A TASK

Thirteen Purposes, Each With Its Own Search

Step	Source	Filters	Limit	Req.
activity_definition	brief	track, activity number	3	●
deliverables	brief	track	2	●
client_context	brief	track, client and materials	4	●
schedule, supervisor_voice, rubric	brief	track, one facet each	1 each	
planted_issues	brief	complication tasks only	3	●
programme_norms	brief	untracked rules	3	
market_responsibilities	job market	role, level, sector, ladder	4	
market_tools, market_collaboration	job market	role, level, sector, ladder	3 each	
culture_signals	employer review	role, level, sector, ladder	2	
brand_voice	brand	brand voice facet	2	

RELAXATION LADDER

Widen Gently, Never Abandon the Role

RUNG 0 | STRICT
role family + job level + sector

RUNG 1 | DROP SECTOR
role family + job level

RUNG 2 | DROP LEVEL
role family only

When it climbs

Only when a market step returns fewer than its minimum hits, usually 2.

Example

Data Engineer in healthcare has few postings at one level, so sector is dropped first.

Transparent

Every climb is written into the response diagnostics.

WORKED EXAMPLE | RECIPROCAL RANK FUSION

Why the Activity 3 Passage Rose From 13th to 1st

ACTIVITY 3 BRIEF PASSAGE	
Semantic rank 13	1 ÷ 73 = 0.01370
Lexical rank 1	1 ÷ 61 = 0.01639
Fused	**0.03009**

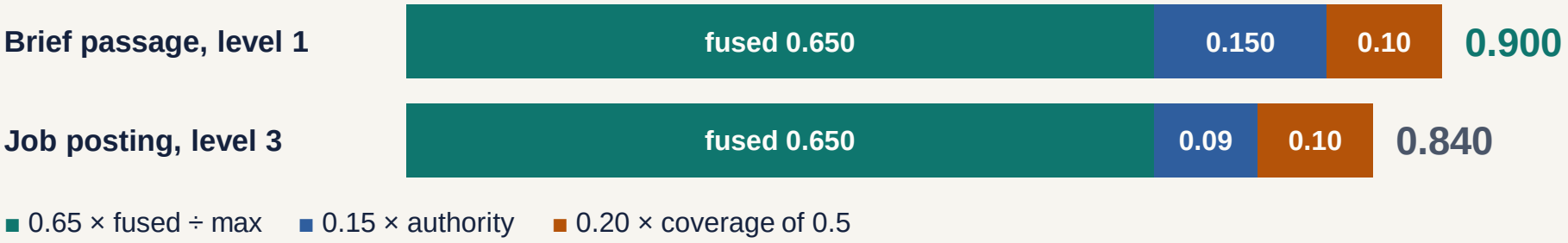
A PASSAGE FOUND BY MEANING ONLY	
Semantic rank 1	1 ÷ 61 = 0.01639
Not in lexical list	0
Fused	**0.01639**

Lesson: agreement between two channels beats first place in one. Exact words like “Activity 3” carried the passage.

Ranks measured in the offline run; k = 60 | Selected architecture | 89

WORKED EXAMPLE | RERANK

Authority Breaks the Tie in Favour of the Brief



Measured: the Activity 3 brief passage finished with a final score of 0.937 in the offline run.

DIVERSITY RULES

Three Filters Keep the Evidence Varied

1

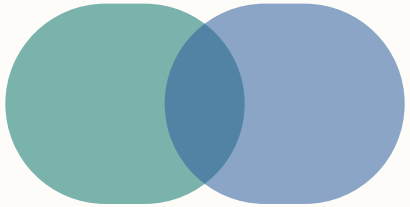
Exact duplicates

hash a1f | ~~hash a1f~~

Same content hash, second copy dropped.

2

Near duplicates



Word overlap (Jaccard) of 0.85 or more: the lower ranked passage is dropped.

3

Employer cap

Firm X | Firm X | ~~Firm X~~

At most 2 passages per employer, so one large recruiter cannot dominate.

SOURCE BALANCED RETRIEVAL | POST /V1/RETRIEVE

Programme Rules Are Never Crowded Out

KNOWLEDGE BASE BY VOLUME



RESULTS FOR LIMIT = 10



Two steps

Programme sources get 60% of the limit; job postings and reviews get the rest.

Filters where they mean something

Level and sector only apply to the market step.

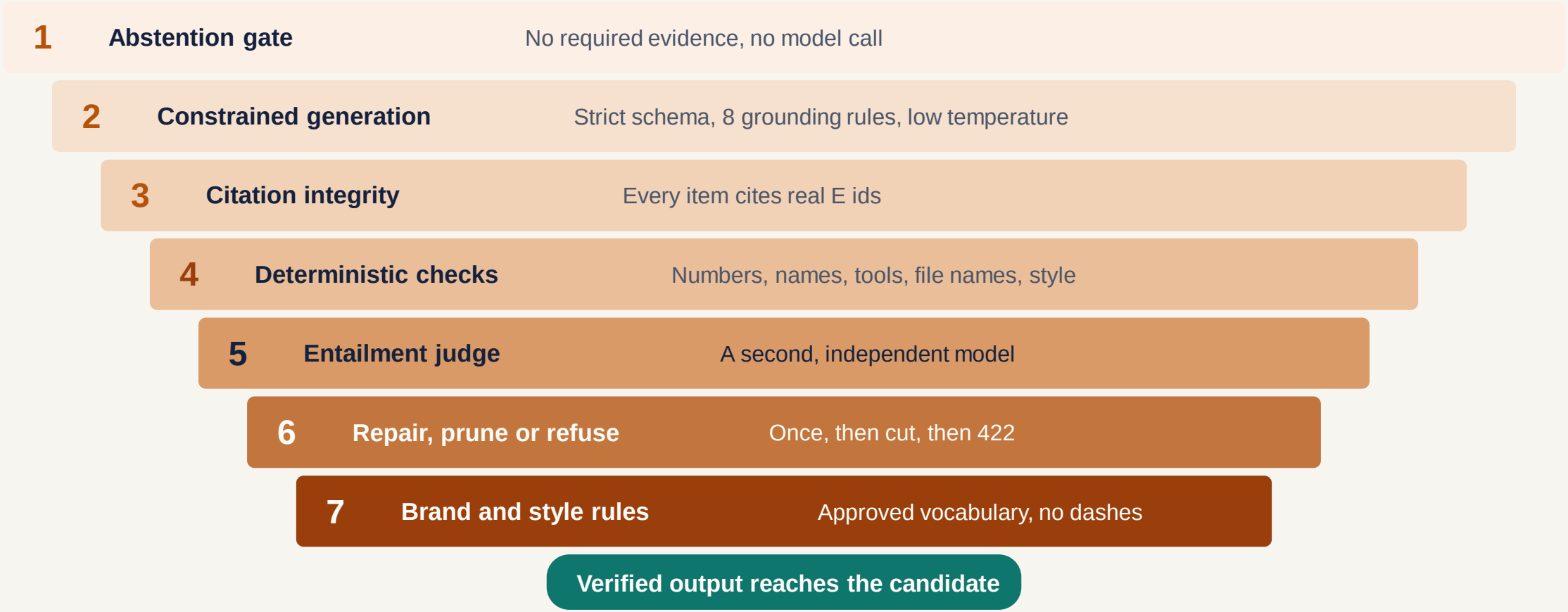
Effect

Retrieval hit rate recovered from 0.875 to 1.00.

Programme share computed from 3,194 chunks: 1,835 job market plus 526 review chunks are market | 92

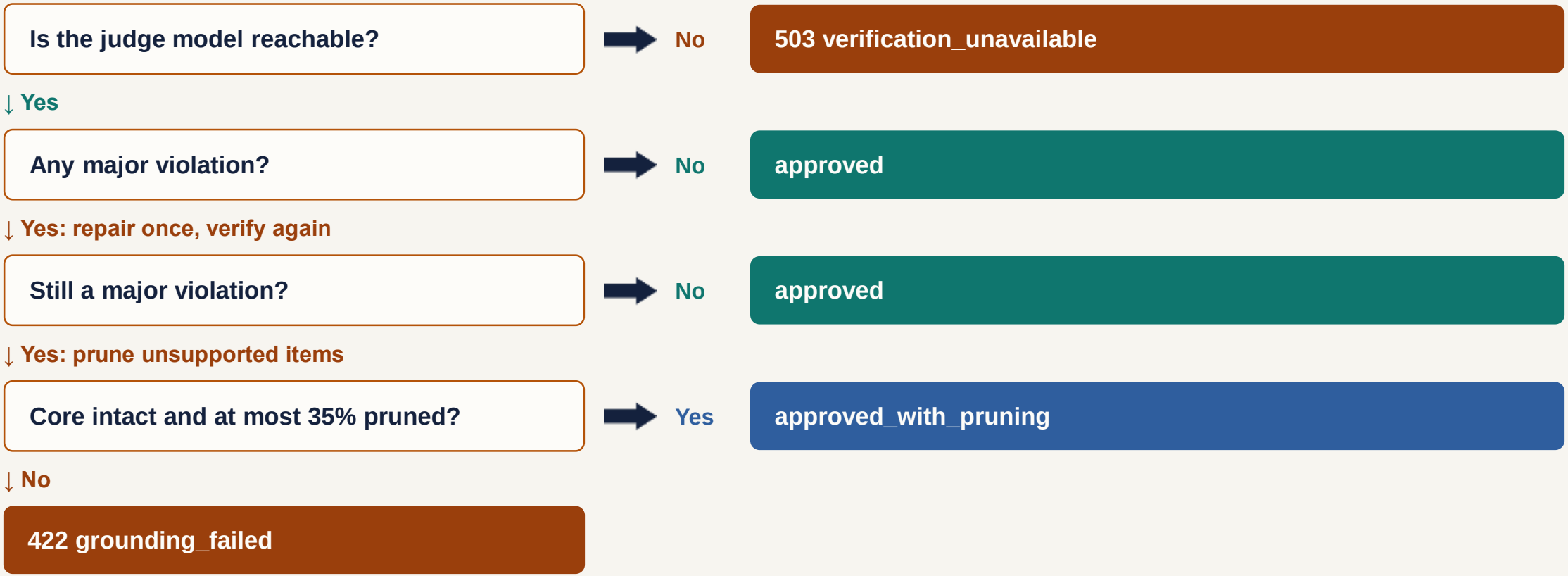
HALLUCINATION DEFENCE

Seven Layers Between the Model and a Candidate



VERDICT TREE

Every Task Ends in One of Four Outcomes



FAIL CLOSED

When Something Breaks, Nothing Unverified Escapes

What fails	What the service does	Platform sees
Required evidence missing	Stops before any model call	422 insufficient_evidence
Generation model down	Tries the fallback model, GPT 4o	502 if both fail
Output breaks the schema	One strict retry, then stops	502 invalid_model_output
Judge model down	Refuses to release unverified text	503 verification_unavailable
Claims unsupported after repair	Prunes, or refuses past 35%	422 grounding_failed
Database unavailable	Readiness probe fails	503 on /health/ready
Audit write fails	Logs the error, still answers	200 with the result

07

The Knowledge Base

Twenty seven source files, how each was read, cleaned and tagged, and the 3,194 chunks that every answer is built from.

SOURCES

27 Files, Seven Source Types, Five Authority Levels

File	Source type	Authority	Chunks
Week One Work Brief, PDF	programme_brief	1	163
Brand Strategy Portfolio, PDF	brand_guideline	2	208
Plain English Report, DOCX	product_overview	2	48
23 market workbooks, XLSX	job_market	3	1,835
same workbooks, review sheets	employer_review	3	526
same workbooks, method sheets	methodology	4	148
Knowledge Graph design, PDF	engineering_reference	5	266
Total			3,194

Engineering reference is excluded from task generation by default; it is available to /v1/retrieve on request.

MARKET COVERAGE

23 Workbooks Across Four Roles and Six Sectors

	Construction	Consulting	Education	Healthcare	IT	General
Cybersecurity	1	1	1	·	1	2
Data Analyst	1	1	1	1	1	1
Data Engineer	·	1	1	1	1	1
Software Developer	1	1	1	·	1	2

Number of workbooks per cell. 21 of 24 combinations covered; gaps: Cybersecurity and Software Developer in healthcare, Data Engineer in construction. Collected 15 September to 22 October 2026.

THE WORK BRIEF

83 Pages, Three Parts, One Repeating Track Pattern

PART A | SECTIONS 1 TO 6

Rules for every participant: environment, weekly rhythm, standards, submission, assessment

PART B | SECTIONS 7 TO 10

Four track briefs, one per discipline

PART C | SECTION 11

Shared appendices and templates

EVERY TRACK BRIEF, AND THE FACET IT BECOMES

1 Supervisor welcome note	supervisor_voice
2 Client and problem	track_client
3 Working materials	track_materials
4 Week at a glance	track_schedule
5 Sequential activities	track_activity
6 Required deliverables	track_deliverables
7 Assessment rubric	track_rubric

THE FOUR TRACKS

Four Fictitious Clients, Four Working Weeks

DA

Data Analyst

Kestrel Lane Home and Garden Ltd

27 planned hours

SE

Software Developer and Engineer

Northwold Couriers Ltd

26 planned hours

CY

Cybersecurity

Ashbourne Valley Credit Union

27 planned hours

DE

Data Engineering

Voltaway Charging Networks Ltd

26 planned hours

Minimum 20 logged hours per participant. Every client, person and dataset is fictitious and created by Phexara Limited.

WEEKLY RHYTHM

The Fixed Events the AI Manager Must Respect

MON	TUE	WED	THU	FRI
Kick off and plan	Discover and build	Build and test	Refine and review	Deliver and present
09:30 Induction call	**by 10:00** Daily update	**by 10:00** Daily update	**by 10:00** Daily update	**by 10:00** Daily update
14:00 Client question window opens	**12:00** Question window closes	**booked slot** Midweek review	**by 12:00** Peer review exchange	**15:30** Showcase
				17:00 Final submission

Source: Work Brief, Section 3, UK time | The knowledge base | 101

ASSESSMENT FRAMEWORK

Five Competencies, Weighted to 100



- Technical Quality, 30
- Problem Solving, 20
- Communication, 20
- Delivery, 15
- Professionalism, 15

Why the RAG needs it

Every task’s competency focus must come from this framework and the track rubric, never from the model’s idea of good work.

Where it lives

The rubric step retrieves the track_rubric facet for every task.

WORKBOOK ANATOMY

One Market Workbook, Sheet by Sheet

SHEET IN THE WORKBOOK	WHAT THE LOADER DOES
Master dataset	Role records and skills facets
Day to day job tasks	responsibilities
Tech stack	tech_stack
Team collaboration	team_collaboration
Work culture ratings	culture_ratings
Reviews and perspectives	employee_review, perspective
Rating methodology, data dictionary	methodology

Sheets are recognised by header aliases, not by sheet name, so a renamed tab still loads. Example: Data Engineer, Consulting, 19 October 2026.

DATA QUALITY

Real Data Is Messy. The Loader Expects It.

Company id typos

NUF 0022 against NUF 002

BRI 001 against BHK 001

Fix: reattach the row by employer name and job title. Separators shown as spaces.

Six date formats in file names

Oct_22nd_2026 · 2026 09 29
1st_of_October_2026
15th_October_2026 · 21st_Oct_2026

Fix: one parser for every pattern.

Header variants

“Company” and “Company Name”; “Location”, “Region” and “Location/Region”; “Job level” and “Job Level”.

Fix: a header alias table.

Inconsistent file names

Software_developer · Data_Analyst__Consulting
Healthcare_Data_Analytics

Fix: case and spacing insensitive role and sector inference, with manifest overrides.

NORMALISATION RULES

Many Spellings In, One Vocabulary Out

JOB LEVEL

entry, junior, beginner	beginner
mid, intermediate	mid_level
senior, advanced	advanced
lead, principal, expert	expert

COMPANY KEY

Lower case, drop Ltd, Limited, PLC, LLP, Inc, Group, UK and The, keep letters and digits.

The Example Group UK Ltd

⬇

example

ROLE AND SECTOR

Inferred from the file name:

4 role families

cybersecurity, data_analyst, data_engineer, software_developer

6 sectors

construction, consulting, education, healthcare, information_technology, general

FACET TAXONOMY

24 Facets Say What Kind of Fact a Chunk Holds

PROGRAMME BRIEF | 10

- programme_rules
- supervisor_voice
- track_client
- track_materials
- track_schedule
- track_activity
- track_deliverables
- track_rubric
- track_stretch
- templates

BRAND, PRODUCT | 4

- brand_voice
- brand_general
- product
- engineering

JOB MARKET | 10

- responsibilities
- skills_experience
- tech_stack
- team_collaboration
- work_culture
- culture_ratings
- employee_review
- employee_perspective
- methodology
- general

Brief facets come from ordered regular expression rules in knowledge_manifest.yaml; market facets come from the sheet kind.

CHUNK ANATOMY

A Chunk Is Text Plus Everything Needed to Find and Cite It

CONTENT

header: document, section path

body: up to 450 tokens

overlap: 60 tokens with the neighbour

content_hash: for dedupe and cache

RETRIEVAL KEYS

embedding: 1,536 numbers

tsv: weighted words, header A, body B

source_type, facet

track, activity number

role families, sectors, level

PROVENANCE

source_file

page_start, page_end

sheet, row_number

company, job title, url

authority: 1 to 5

Design choice: the header is repeated inside every chunk, so a passage still makes sense on its own.

CHUNK EXAMPLE | PROGRAMME BRIEF

Data Analyst, Activity 3, as the Service Stores It

Activity 3: Data profiling and quality audit

Scheduled Monday and Tuesday · Effort 3.0 hours · Depends on Activity 2 · Produces D2 Data Quality Log

Objective Measure the quality of every dataset systematically before changing anything, so that every later decision is grounded in evidence.

What you must do, in order 1. For each file, record the number of rows and columns, the data type of each field, the percentage of nulls ... 2. Check the uniqueness of every intended key ... 3. Check referential integrity ...

... continues to step 7 and the completion criteria

```
# metadata
source_type: programme_brief
authority: 1
facet: track_activity
track: DA
activity_number: 3
page_start: 24
section_path: 7 Track 1 ...
```

Text from the Work Brief, page 24. Bold labels such as “Objective” are preserved as paragraph labels by the parser.

CHUNK EXAMPLE | JOB MARKET

One Real Posting Becomes a Responsibilities Chunk

Accenture · Data and AI Engineer · Bristol

Data Engineer workbook, consulting, collected 19 October 2026

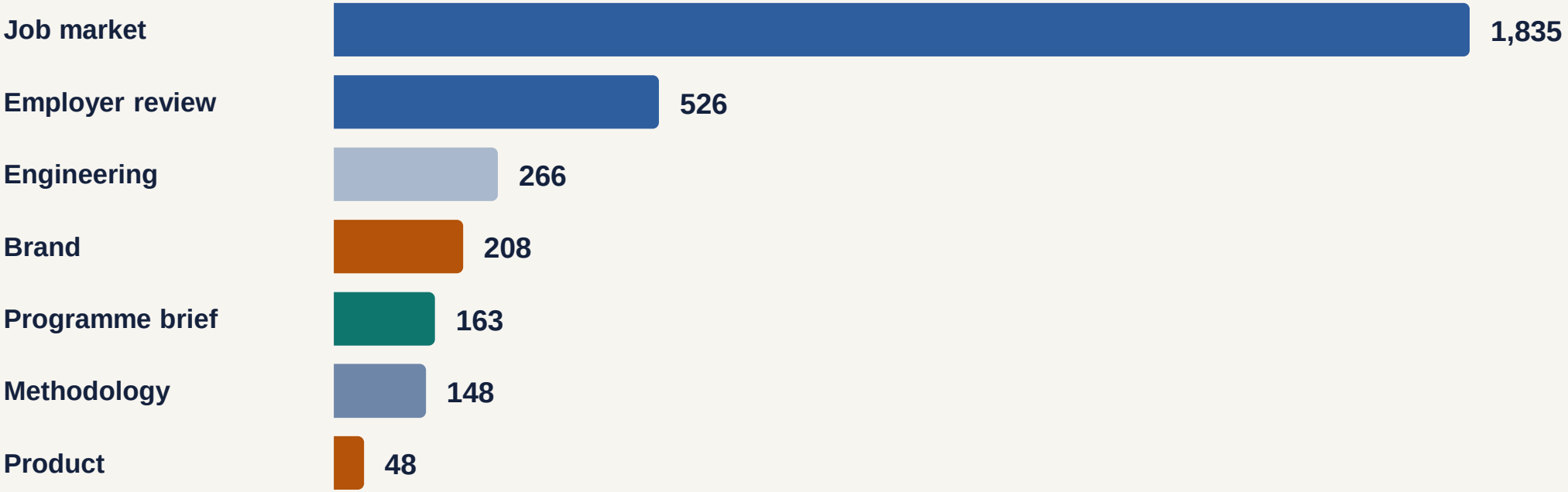
Day to day responsibilities Build and operate cloud data and platform solutions; develop and maintain scalable batch and streaming pipelines; ingest, validate, normalise, cleanse and transform data; provision and manage cloud environments using IaC; apply data modelling; implement data quality, lineage, observability and monitoring ...

```
# metadata
source_type: job_market
authority: 3
facet: responsibilities
role_families: [data_engineer]
sectors: [consulting]
job_level: mid_level
company_key: accenture
sheet, row_number
```

Excerpt from the workbook row. The source spelling of the level was normalised to mid_level; the cap of 2 per employer applies to Accenture.

INGESTION RESULT

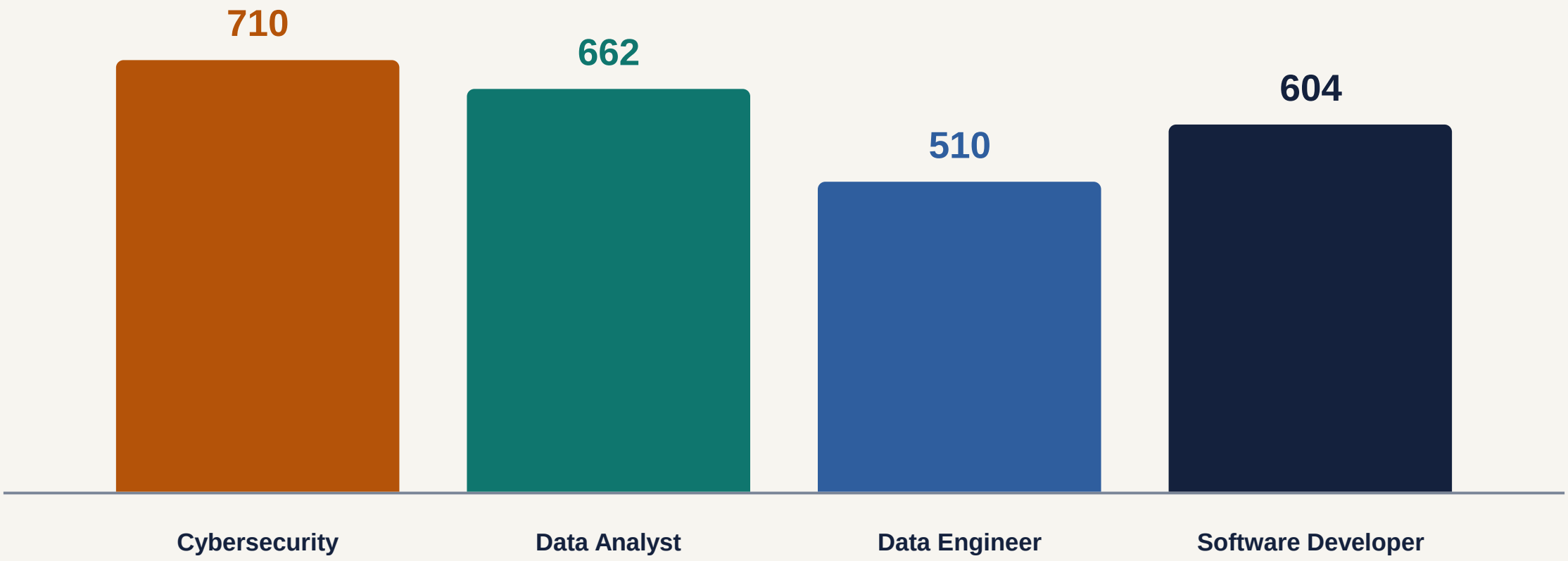
3,194 Chunks by Source Type



The brief is only 5% of chunks yet carries the highest authority. That imbalance is why retrieval is planned per purpose and source balanced.

INGESTION RESULT

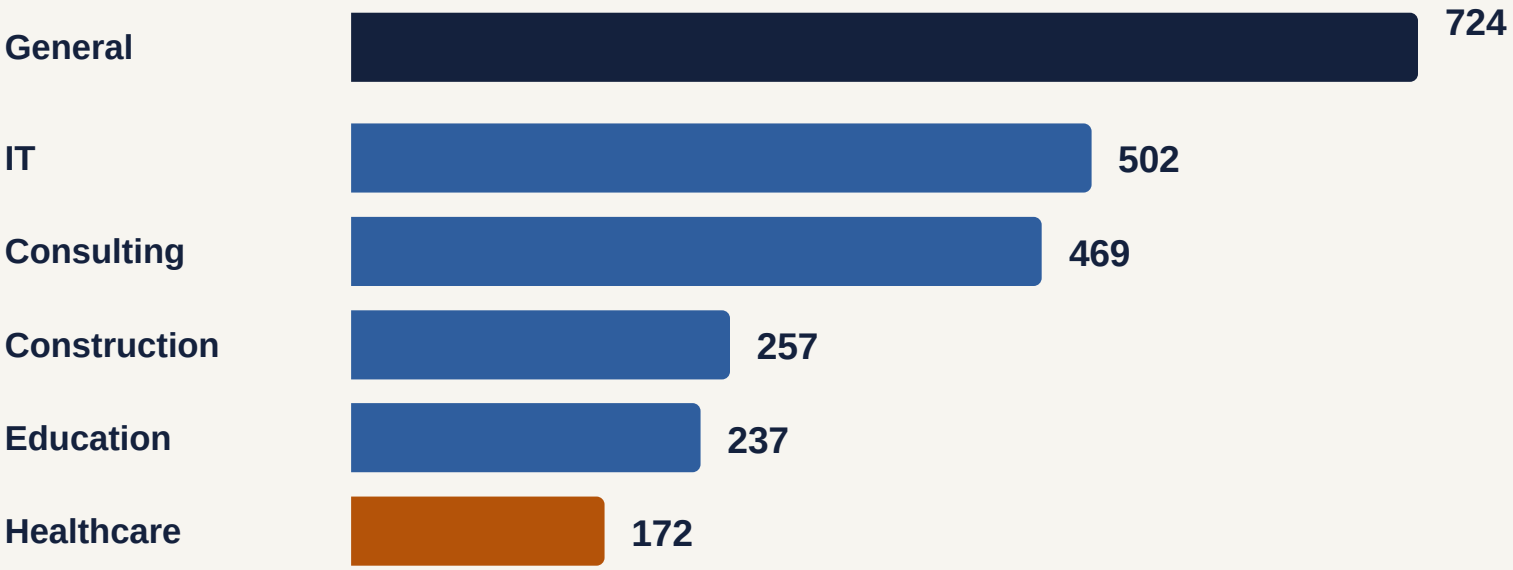
Market Chunks by Role Family Are Well Balanced



Chunks tagged with each role family | The knowledge base | 111

INGESTION RESULT

Market Chunks by Sector



Thinnest sector

Healthcare has no cybersecurity or software developer workbook; the relaxation ladder covers these requests.

Next collection

Prioritise the three empty cells in the coverage matrix.

48 ACTIVITIES

Four Tracks, Twelve Activities Each

DA Kestrel Lane	SE Northwold	CY Ashbourne	DE Voltaway
1 Induction, set up	1 Induction, tooling	1 Induction, evidence	1 Induction, set up
2 Questions, plan	2 Requirements, stories	2 Scoping, plan	2 Source discovery
3 Profiling, quality audit	3 System design	3 Timeline reconstruction	3 Pipeline architecture
4 Cleaning, transformation	4 Repository, CI skeleton	4 Impact and exposure	4 Local warehouse
5 Data model, KPIs	5 Booking endpoints	5 Containment, recovery	5 Bronze ingestion
6 Sales and margin	6 Status lifecycle	6 Posture vs UK guidance	6 Silver cleaning
7 Promotions analysis	7 Public tracking	7 Risk register	7 Data quality tests
8 Loyalty segmentation	8 Auth and roles	8 Improvement roadmap	8 Gold model
9 Dashboard build	9 Automated testing	9 Control procedures	9 Revenue marts
10 Insight report	10 Security review	10 Awareness briefing	10 Orchestration
11 Executive presentation	11 API documentation	11 Incident report	11 Lineage, runbook
12 Reflection, handover	12 Release, reflection	12 Showcase, reflection	12 Demo, reflection

Activity titles shortened from the Work Brief, Part B

BRAND VOICE RULES

The Words the AI Manager Never Uses

~~VEEP~~	**Vet2Employ**	~~chatbot~~	**AI Manager**
~~certificate~~	**Verified Profile**	~~grade~~	**score**
~~game~~	**simulation**	~~guaranteed hire~~	**verified evidence**
~~placement fee~~	**Hire Success Fee**	dashes	**commas, colons, full stops**

Context aware: a TLS certificate in a cybersecurity task is a technical object and is allowed. Source: Brand Strategy Portfolio, chapter 4 and Appendix E; config/brand_voice.yaml.

INGESTION REPORT

No Failures. Four Issues Resolved. One for Humans.

- ✓ Column and sheet naming unified across all four batches
- ✓ Mistyped company ids reattached by employer and title
- ✓ Missing Source URL and Niche columns filled from the Master sheet
- ✓ Extra culture rows kept as company culture evidence

NEEDS REVIEW

Healthcare Data Analytics workbook: the day to day row for company GSK 001 names a different job title from its Master record.

The service did not guess. It kept the row as its own record and flagged it for the research team to confirm.

08

The Codebase, Step by Step

Every module in the order data flows through it: configuration, domain, ingestion, storage, retrieval, generation, verification, the API and operations. Each slide says what it does in plain words, then shows how.

REPOSITORY MAP

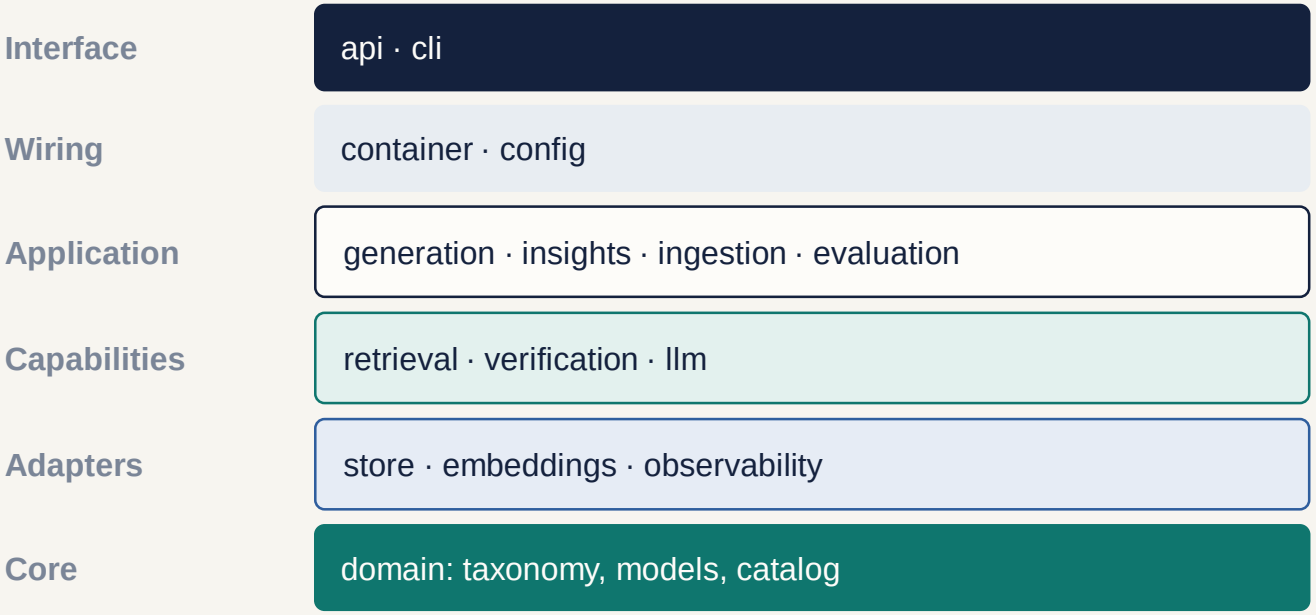
Where Everything Lives

```
vet2employ_rag/
src/v2e_rag/
api/ domain/ ingestion/
embeddings/ store/ retrieval/
llm/ verification/ generation/
insights/ observability/ evaluation/
config.py container.py cli.py
config/ migrations/ eval/
tests/ docs/ knowledge_base/raw/
Dockerfile compose.yaml Makefile
pyproject.toml .github/workflows/
```

src/	the service, 12 packages
config/	tracks, manifest, brand voice
migrations/	PostgreSQL schema
eval/	24 golden cases, 13 scenarios
tests/	96 automated tests
docs/	6 guides for developers
Docker	image and local stack

MODULE LAYERS

Dependencies Point Downward Only



For non engineers

Like a building: the foundations (domain) never depend on the furniture (API). Change the furniture freely.

For engineers

8,370 lines across 12 packages. Swappable adapters behind protocols; no import cycles.

TECHNOLOGY STACK

Mainstream, Supported, Boring on Purpose

Runtime

Python 3.11+

FastAPI, Uvicorn

Pydantic v2

Typer CLI

Data

PostgreSQL 15+

pgvector, HNSW

psycopg 3, pool

full text search

Models

OpenRouter API

httpx, tenacity

Claude Sonnet 4.5

Claude Haiku 4.5

Documents

pdfplumber

python docx

openpyxl

rapidfuzz

Operations

structlog JSON logs

Prometheus metrics

Docker, compose

Quality

pytest, respx

ruff

mypy

Delivery

GitHub Actions CI

Makefile targets

No lock in

Models are a config value; any OpenRouter model can be swapped in.

DESIGN PATTERNS

Six Patterns That Keep the Code Honest

Ports and adapters

VectorStore, Embedder and ChatModel are protocols; PostgreSQL and memory stores both fit.

Composition root

container.py builds every object once from settings. No hidden globals.

Configuration as data

Tracks, facets and brand rules in YAML, editable without code.

Typed contracts

Pydantic models at every boundary, extra fields forbidden.

Fail closed

Typed exceptions map to precise HTTP errors; no silent fallbacks.

Idempotent pipelines

Ingestion keyed by file hash; rerunning changes nothing.

CODE WALK 01 | CONFIG.PY

Every Dial Is an Environment Variable

One settings object. Each value can be changed per environment without touching code.

Production guard

Refuses to start in production without API keys, or with the offline hashing embedder.

Shared key

Accepts the platform's existing OPENROUTER_API_KEY; data collection set to deny.

```
# .env, prefix V2E_
GENERATION_MODEL=<Claude Sonnet 4.5>
VERIFIER_MODEL=<Claude Haiku 4.5>
EMBEDDING_DIMENSIONS=1536
CHUNK_MAX_TOKENS=450
RRF_K=60
EVIDENCE_TOKEN_BUDGET=7000
MAX_EVIDENCE_ITEMS=24
VERIFICATION_MAX_REPAIR_ATTEMPTS=1
STYLE_FORBID_DASHES=true
PII_REDACTION=true
```

CODE WALK 02 | DOMAIN/TAXONOMY.PY

The Shared Vocabulary of the Whole System

SourceType

7 values

programme_brief, brand, product, job_market, employer_review, methodology, engineering

SOURCE_AUTHORITY

1 to 5

Lower is stronger. The brief is 1, engineering notes are 5.

Facet

24 values

What kind of fact a chunk holds.

RoleFamily, Sector, JobLevel, TrackCode

4, 6, 4, 4 values

Plus normalise_job_level and company_key.

Why it matters: enums make a typo a startup error, not a silent empty search.

CODE WALK 03 | DOMAIN/MODELS.PY

Six Objects Carry Data Through the Whole Pipeline

SourceDocument
one file, its hash and type

➡

RoleRecord
one job role from a workbook

➡

Chunk
searchable passage with 24 metadata fields

➡

SearchHit
ranks and scores per channel

➡

EvidenceItem
a hit with an E id and citation

➡

EvidencePack
items, diagnostics, render()

estimate_tokens
max(words × 1.33, characters ÷ 4) + 1. No external tokenizer needed.

citation_label
Document, section, page; or employer, title, workbook.

CODE WALK 04 | DOMAIN/CATALOG.PY AND CONFIG/*.YAML

Business Knowledge Lives in Three YAML Files

tracks.yaml

The four tracks: name, role family, heading patterns that find each track in the brief, manager title, retrieval keywords.

Deliberately holds no names of people or clients. Those are read from evidence.

knowledge_manifest.yaml

Which file is which source type and layout, workbook overrides, and ordered facet rules as regular expressions.

A new document needs one entry, not new code.

brand_voice.yaml

Banned terms with a reason, a replacement and context exceptions, plus phrases that would break character.

Marketing can update vocabulary safely.

load_catalog() reads all three at startup into frozen dataclasses; a malformed file stops the service before it serves a request.

CODE WALK 05 | INGESTION/FILENAME_METADATA.PY

A File Name Becomes Role, Sector and Date

Healthcare_Data_Engineering_1st_of_October_2026.xlsx



ROLE_FAMILIES

data_engineer

pattern: data\s*engineer

SECTORS

healthcare

pattern: healthcare, health care, NHS

COLLECTED_ON

1 October 2026

ordinals and month names parsed

No sector in the name means general. A manifest override wins over any inference. Upload prefixes are stripped first.

CODE WALK 06 | INGESTION/EXCEL_LOADER.PY, PART 1

Sheets Are Recognised by What They Contain

CLASSIFY_SHEET DECISION ORDER

1 name says methodology or dictionary

2 name says perspective

3 name says master

4 has an employee review column

5 ratings but no role content

6 has a job title: a topic sheet

```
# HEADER_ALIASES
"company name" → company
"location/region" → location
"skills required" → skills
"job level" → job_level
```

Clean values

N/A and blanks become empty. Ratings outside 1 to 5 are dropped, never estimated.

CODE WALK 07 | INGESTION/EXCEL_LOADER.PY, PART 2

Topic Sheets Merge Into One Record per Role



Ratings rows that duplicate a Master record are merged; ratings for employers without a Master record are kept as company culture records.

CODE WALK 08 | INGESTION/DOCUMENT_PARSER.PY, PART 1

Reading a PDF’s Structure From Its Fonts

7. Track 1: Data Analyst

7.5 Sequential activities

Activity 3: Data profiling

body text, size 1.0×

1 Find the body size

The most common font size is body text.

2 Rank structural sizes

Larger or bold sizes used by 3 or more headings become levels 1 to 6.

3 Weight fallback

Decks with bold body size headings, like the brand portfolio, switch to weight rules.

CODE WALK 09 | DOCUMENT_PARSER.PY, PART 2

Tables Become Self Describing Sentences

Event	When	Duration
Induction call	Monday 09:30	45 minutes
Peer review	Thursday by 12:00	60 minutes



Event: Induction call | When: Monday 09:30 | Duration: 45 minutes

Event: Peer review | When: Thursday by 12:00 | Duration: 60 minutes

Why: each line carries its own column names, so a single row still embeds, matches “Monday 09:30” and cites cleanly.

CODE WALK 10 | DOCUMENT_PARSER.PY, PART 3

Remove Page Furniture, Keep Meaningful Labels

~~PHEXARA LIMITED | Work Brief~~

Objective:

Measure the quality of every dataset ...

You are done when:

...

~~Page 24 of 83~~

Boilerplate

Only lines in the top and bottom 9% of a page that repeat on 30% of pages are removed.

Labels

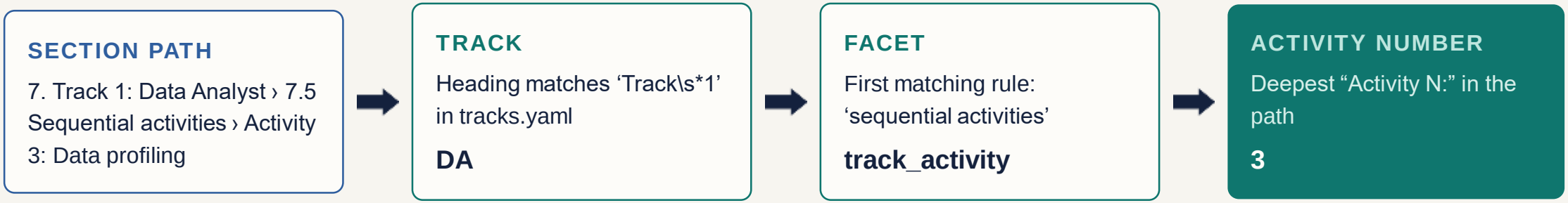
Short bold lines such as “Objective” repeat in every activity, yet sit in the body, so they are kept and end with a colon.

A bug found and fixed

An early version removed these labels everywhere; limiting removal to margins fixed it.

CODE WALK 11 | INGESTION/CHUNKING.PY, PART 1

Every Section Is Tagged With Where It Belongs



The activity number is what lets a step ask for exactly Activity 3 of the Data Analyst track and nothing else.

CODE WALK 12 | CHUNKING.PY, PART 2: PACK_PARTS

Windows of 450 Tokens, Overlapping by 60



Whole parts first

Paragraphs and list items are never cut mid sentence.

Oversize parts

Split on sentence boundaries before packing.

Overlap carry

Trailing parts up to 60 tokens repeat in the next chunk.

CODE WALK 13 | CHUNKING.PY, PART 3: ROLECHUNKER

One Role Becomes Several Focused Facet Chunks

RoleRecord

Accenture, Data and AI Engineer, mid level, consulting



- **responsibilities** Day to day responsibilities
- **skills_experience** Skills and experience required
- **tech_stack** Frameworks, languages, databases, cloud, tools
- **team_collaboration** Team structure and collaboration
- **culture_ratings** Ratings with review counts

Every chunk repeats the header: Role | Employer | Niche | Location | Level | Role family | Market sector | Collected

CODE WALK 14 | INGESTION/PIPELINE.PY

Run It Twice and Nothing Changes

Compute file sha256	Same hash as stored and no force flag? Mark unchanged and skip.
Parse into units	One document, its chunks and its role records per unit.
Dedupe reference sheets	The methodology sheet in every workbook is stored once.
Embed in batches of 64	Cached by content hash, so unchanged text is never paid for twice.
replace_document	Old chunks out, new chunks in, in one transaction.
Per file isolation	One bad file is reported as failed; the rest still ingest.

CODE WALK 15 | EMBEDDINGS/BASE.PY

Three Embedders Behind One Interface

OpenRouterEmbedder

Batches of 64, data collection denied.

Retries 408, 409, 425, 429 and 5xx with exponential jitter.

Checks dimensions match settings; vectors normalised to unit length.

Query cache of 2,048 entries.

CachedEmbedder

Wraps any embedder. Looks up model plus content hash in embedding_cache first.

Only new texts are sent; repeated texts in one call are embedded once.

HashingEmbedder

Deterministic, offline, free. For tests and laptops only.

Settings refuse it in production.

Model: text embedding 3 small, 1,536 dimensions, matching the platform’s existing artefacts table.

CODE WALK 16 | STORE/BASE.PY

One Storage Contract, Two Implementations

```
class VectorStore(Protocol):
open, close, health
get_document, list_documents
replace_document, delete_document
semantic_search(vector, filters, n)
lexical_search(query, filters, n)
get_chunks, role_records, stats
cached_embeddings, cache_embeddings
record_audit, record_ingestion_run
```

SEARCHFILTERS

OR within a field, AND across fields. Empty means no constraint.

- source_types
- facets
- role_families
- sectors
- job_levels
- tracks
- activity_numbers
- exclude_source_types
- include_untracked
- only_untracked

Same filters, both channels

Applied inside the vector and keyword queries, never after.

CODE WALK 17 | STORE/PGVECTOR_STORE.PY

The Two SQL Queries That Do the Searching

```
# semantic
SELECT ...,
1 − (embedding <=> q) AS similarity
FROM kb_chunks WHERE {filters}
ORDER BY embedding <=> q
LIMIT 40
```

```
# lexical
SELECT ...,
ts_rank_cd(tsv, q, 32) AS rank
FROM kb_chunks,
to_tsquery('english', terms) q
WHERE tsv @@ q AND {filters}
ORDER BY rank DESC LIMIT 40
```

ef_search 80

HNSW recall tuned per query, iterative scan where supported.

Exact fallback

A selective filter that under returns triggers an exact scan.

Safe terms

Query words are sanitised and OR joined, at most 32.

All values bound as parameters; filters compiled by _filter_sql | The codebase | 137

Indexes Chosen for Every Query Path

Index	Type	Serves
embedding	HNSW, m 16, ef 64	semantic search, cosine
tsv	GIN	keyword search; header weighted A, body B
role_families, sectors	GIN	array overlap filters on market steps
source_type, facet	btree	every planned step
track, job_level	btree	brief steps and the level filter
document_id, content_hash	btree	replace on ingest, dedupe
audit created_at; operation, status	btree	audit review and dashboards

The tsv column is generated by the database, so keyword data can never drift from the text. Compatible with PostgreSQL 15+ and AlloyDB.

CODE WALK 19 | STORE/MEMORY_STORE.PY AND LEXICAL.PY

The Same Logic, No Database Needed

Why it exists

Tests, laptops and evaluation runs exercise identical hybrid retrieval without PostgreSQL.

Semantic

Exact cosine with numpy over filtered chunks.

Snapshot

Saved to var/memory_store, reloaded on start.

BM25 LEXICAL CHANNEL

score = Σ idf(t) × tf × (k1 + 1) ÷ (tf + k1 × (1 − b + b × len ÷ avg))

k1 = 1.4, b = 0.75, English stopwords removed.

Measured: the memory backend reached a 1.00 hit rate on the golden set.

CODE WALK 20 | RETRIEVAL/PLANNER.PY

Three Planners for Three Kinds of Request

plan_task

Up to 13 steps: brief purposes, programme norms, four market steps with ladders, brand voice.

No track? Programme norms and market responsibilities become required.

plan_manager_reply

brief_answer (5), supervisor_voice (1), market_context (3), brand_voice (1).

Similarity threshold on, so weak matches cannot pass as answers.

plan_retrieve

Programme step takes 60% of the limit, market step the rest.

Explicit source types collapse it to one step.

Key idea: the planner is plain Python. It never calls a model, so plans are free, fast and identical every time.

CODE WALK 21 | RETRIEVAL/HYBRID.PY

HybridRetriever.search in Five Lines

```
semantic, lexical = await gather(
store.semantic_search(vector, filters, 40),
store.lexical_search(query, filters, 40))
if threshold:
semantic = [s for s in semantic if s.score >= 0.20]
fused = reciprocal_rank_fusion(semantic, lexical, k=60)
return rerank(fused, query, limit)
```

Parallel channels

Both queries run at the same time.

Lexical only guard

With the threshold on, a keyword only hit needs a real score.

Measured

Latency recorded in a Prometheus histogram.

CODE WALK 22 | RETRIEVAL/EVIDENCE.PY

EvidenceBuilder: Execute, Check, Assemble

execute

Embed unique queries; run every step in parallel; climb the ladder per step until min hits.

check

Any required step empty? Raise InsufficientEvidenceError with the missing names and diagnostics.

assemble

Required first, round robin, dedupe by id and hash, stop at 7,000 tokens or 24 items.

Diagnostics returned: each step’s purpose, hit count, required flag, relaxation level, total tokens and items.

CODE WALK 23 | LLM/OPENROUTER.PY, PART 1

One Request to OpenRouter, Built Carefully

```
POST /chat/completions
{ "model": <Claude Sonnet 4.5>,
"models": [sonnet, <GPT 4o>],
"temperature": 0.2,
"max_tokens": 4000,
"provider": {
"data_collection": "deny",
"require_parameters": true },
"response_format": {
"type": "json_schema", "strict": true } }
```

models list

OpenRouter falls back to GPT 4o if Sonnet is unavailable.

data_collection deny

Only providers that do not train on prompts are used.

require_parameters

Only routes to providers that honour the strict schema.

CODE WALK 24 | LLM/OPENROUTER.PY, PART 2

Resilience: Retries, Mode Fallback, Clean Errors

Retries

Up to 3 retries with exponential jitter, capped at 30 seconds.

Only on timeouts, transport errors, 408, 409, 425, 429 and 5xx.

Output mode ladder

- json_schema
- json_object
- prompt only

Next mode only if a provider rejects the parameter.

Clean errors

LLMError for HTTP or transport problems; LLMOutputError for unusable JSON.

extract_json tolerates code fences; usage and model are recorded.

CODE WALK 25 | LLM/SCHEMAS.PY

The Shape of a Generated Task

GeneratedTask
title, sender_role, priority P0 to P2
message: what the candidate reads
summary, context
objectives, steps (at least one each)
deliverables with file name and due
tools, collaboration, complications
competencies, schedule, open_questions

GroundedItem

text: one self contained statement

evidence_ids: ["E3", ...]

Used for every factual field.

ManagerReply

reply, intent, claims, needs_human_review, review_reason

JudgeVerdict

SUPPORTED, PARTIAL, UNSUPPORTED, NOT_A_CLAIM

CODE WALK 26 | LLM/PROMPTS.PY, VERSION 2026.10.1

The Eight Non Negotiable Grounding Rules

1. Evidence is the only source of facts.
2. Every item cites real evidence ids.
3. Missing facts go to open_questions, never filled.
4. Names and files copied exactly; roles not invented names.
5. The programme brief overrides every other source.
6. Market data adds realism only; never a real employer as client.
7. Every number must appear in cited evidence.
8. Due points only from the schedule; otherwise null.

Rules ask; the verifier enforces. Every audit row records the prompt version that produced it.

CODE WALK 27 | VERIFICATION/CLAIMS.PY

Break the Output Into Checkable Claims

GeneratedTask

one JSON object, many fields



summary core

objectives[0], steps[0..n] core

context[i], tools[i], collaboration[i]

deliverables[i], acceptance_criteria[j]

competencies[i], schedule

message sentences kind = sentence

Item claims

Checked against their own cited evidence.

Sentence claims

The friendly message is checked against everything the task cites.

Core flag

Losing a core claim means refuse, not prune.

CODE WALK 28 | VERIFICATION/CHECKS.PY

Six Deterministic Checks, No Model Involved

Check	Rule	Severity	Method
citations	Every item cites at least one id that exists in the pack	major	set lookup
numbers	Every number appears in the cited evidence or the request	major	regex
entities	Every capitalised name appears in the evidence or request	major	regex
tools	A named tool appears in its cited evidence	major	fuzzy, 88
file names	Every V2E_ file name appears verbatim in the evidence	major	regex
style	No banned brand terms, no character breaks, no dashes	minor	config

Major findings trigger repair or pruning. Minor findings are fixed by the final brand pass. Names, numbers and files are where invention hurts most.

WORKED EXAMPLE | ENTITY CHECK

Catching an Invented Colleague

MODEL WROTE

“Share your log with Sarah Jenkins in finance before the midweek review.”

CHECK_ENTITIES

“Sarah” and “Jenkins” are capitalised mid sentence

searched in evidence and request: **not found**

entity · major · [Jenkins, Sarah]

AFTER REPAIR

“Share your log with your track supervisor before the midweek review.”

Rule 4 in action: when evidence gives a role but no name, refer to the role. Allowed words include the brand, track names, and the names the platform supplies.

WORKED EXAMPLE | NUMBER CHECK

Catching a Deadline That Moved by an Hour

EVIDENCE E4 SAYS

Event: Final submission | When: Friday by **17:00** UK time

MODEL WROTE, CITING E4

“Upload every deliverable by Friday 16:00.”

number · major · not in cited evidence: [16:00]

Repaired: “by Friday 17:00 UK time”

One hour wrong costs a candidate a late penalty. A language model judge may let it pass; a regular expression never does.

CODE WALK 29 | VERIFICATION/ENTAILMENT.PY

A Second, Independent Model as Fact Checker

Different model

Claude Haiku 4.5, fallback GPT 4o mini. Not the writer grading itself.

Only the cited passages

Each claim is judged against the evidence it cites, nothing else.

Batches of 20, temperature 0

No citations at all means UNSUPPORTED without a call.

SUPPORTED	every detail stated
PARTIAL	core yes, one detail no
UNSUPPORTED	absent or contradicted
NOT_A_CLAIM	greeting, sign off

A judge failure is returned as an error, never as “approved”.

CODE WALK 30 | VERIFICATION/VERIFIER.PY

The Verifier Produces One Report

VERIFY() RUNS, IN ORDER

- check_citations
- check_numbers
- check_entities
- check_tools, check_file_names
- check_style
- **entailment judge**

```
# VerificationReport
verdict: "approved"
groundedness: 1.0
claims_total, claims_factual
claims_supported
violations: [ ... ]
checks_run: [7 names]
entailment_model
repair_attempts, pruned
```

groundedness = supported factual claims ÷ factual claims. Greetings do not count either way.

CODE WALK 31 | GENERATION/SERVICE.PY: GENERATE_TASK

The Orchestrator That Ties Every Module Together

```
plan = planner.plan_task(request)
pack = await evidence.execute(plan)
messages = task_messages(request, pack)
task = await generate_validated(messages)
report = await verify(task, pack)
while report.major and attempts < 1:
task = await repair(task, report.problems())
report = await verify(task, pack)
task = prune_or_refuse(task, report)
task = finalise_task(task, brand)
await audit(...)
```

Every exit audited

insufficient_evidence, verification_unavailable, rejected and approved all write a row.

Complication guard

A complication task with no supported complication is refused.

CODE WALK 32 | GENERATION/SERVICE.PY: MANAGER_REPLY

Replies Stay in Character, Even When Unsure

Last 12 turns

Conversation history is trimmed and each turn redacted before it reaches the model.

Answer evidence required

No brief or programme hit means an in character abstention and human review, at zero model cost.

Honest about being AI

If the candidate sincerely asks, one sentence says it is an AI persona; the style check allows it only then.

Safer fallback

If pruning would break the reply, it becomes the abstention instead of a refusal error.

CODE WALK 33 | GENERATION/PRUNING.PY

Surgical Removal of Unsupported Content

Can be cut

a context item

a tool or collaboration line

one acceptance criterion

a whole deliverable

the schedule, set to null

up to 30% of message sentences

Cannot be cut

the summary

every objective

every step

Losing these raises PruningError, so the service refuses.

Then finalise

Brand replacements as a last resort.

En and em dashes turned into commas.

Applied to the task and to the reply.

CODE WALK 34 | GENERATION/PII.PY

Personal Data Never Leaves for a Model Provider

CANDIDATE WRITES

“Email me at jo.bloggs@example.com or call 07700 900123.”



MODEL RECEIVES

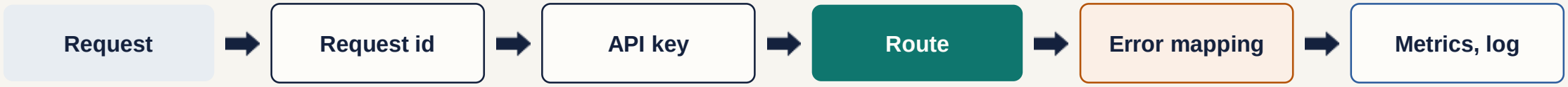
“Email me at [EMAIL REDACTED] or call [PHONE REDACTED].”

email | card or account number | National Insurance number | UK phone | sort code | postcode | keys and passwords

Mirrors the Work Brief data handling rule, Section 4.2. Counts returned in diagnostics | 156

CODE WALK 35 | API/APP.PY

Every Request Passes Through the Same Middleware



Lifespan

Builds the container on start; closes pool and HTTP clients on stop.

Typed error body

error, message, details, plus an X Request ID header. The same shape for every failure.

OpenAPI

Generated docs at /docs with every request and response model.

CODE WALK 36 | API/ROUTES

Fourteen Endpoints, Four Audiences

Method	Path	Purpose	Key
POST	/v1/tasks/generate	grounded work item	service
POST	/v1/manager/reply	in character reply	service
POST	/v1/retrieve	evidence and grounding prompt	service
POST	/v1/verify	check text from elsewhere	service
GET	/v1/catalog	tracks and taxonomy	service
GET	/v1/insights/role_profile	market profile, no model	service
GET, POST, DELETE	/v1/admin/documents	list, upload, delete	admin
POST, GET	/v1/admin/ingest	start and track ingestion	admin
GET	/health/live, /health/ready	probes	none
GET	/metrics	Prometheus	network

CODE WALK 37 | API/SECURITY.PY

Two Keys, Compared in Constant Time

Service key

Sent by the platform in the X API Key header. Opens generation, retrieval and insights.

Wrong or missing: 401.

Admin key

Separate list. Opens uploads, deletes and ingestion. An admin key also works as a service key.

Wrong or missing: 403.

Hardening

hmac.compare_digest resists timing attacks.

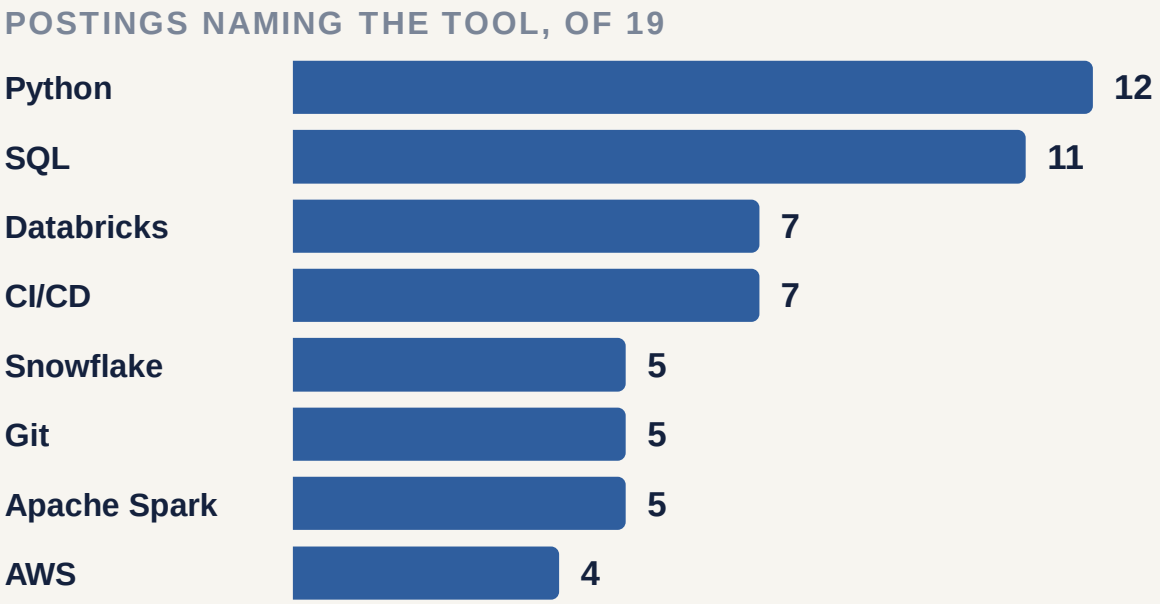
Keys rotate by listing old and new together.

No keys outside development: 503, never open.

Uploads are capped at 50 MB. CORS origins are an explicit allow list, empty by default.

CODE WALK 38 | INSIGHTS/ROLE_PROFILE.PY

Real Result: Data Engineer, Consulting



19 / 13

postings / employers

3.87

mean culture rating, 18 employers, range 2.9 to 4.3

No model involved

Pure counting over role records. Free and exact.

GET /v1/insights/role_profile?role_family=data_engineer&sector=consulting | 160

Nine Metrics That Tell You the Truth Rate

Metric	Type	Answers the question
v2e_http_requests_total	counter	How much traffic, which status codes?
request latency	histogram	How fast per route?
retrieval latency	histogram	Is search slowing down?
llm requests	counter	Which model actually answered, any errors?
v2e_llm_tokens_total	counter	What will the bill be?
v2e_embedding_requests_total	counter	Is the cache working?
verification outcomes	counter	Approved, pruned, rejected, unavailable?
unsupported claims	counter	Which check catches the most?
abstentions	counter	Where is the knowledge base thin?

CODE WALK 40 | CLI.PY

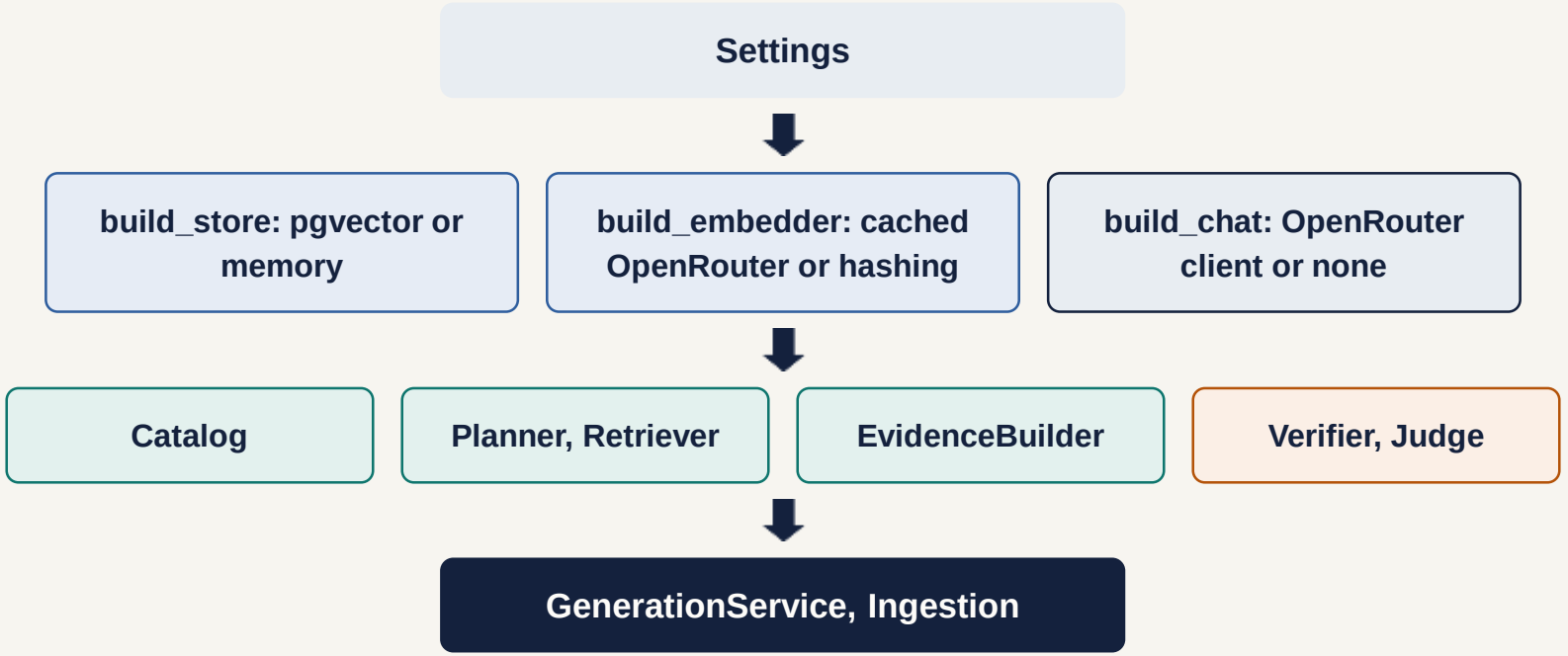
Eight Commands for Developers and Operators

v2e migrate	create schema
v2e ingest	load knowledge
v2e stats	chunk counts
v2e serve	run the API
v2e search	try retrieval
v2e generate_task	one task end to end
v2e eval_retrieval	golden set metrics
v2e eval_generation	13 scenarios

Command names shown as the Python functions; Typer exposes each one. The Makefile wraps them: make up, make ingest, make test.

CODE WALK 41 | CONTAINER.PY

Wiring Everything Together, Once



Tests build the same container with fakes. Without an OpenRouter key, retrieval still works and generation returns a clear 503.

CODE WALK 42 | DOCKERFILE AND COMPOSE.YAML

One Command Brings Up the Whole Stack

IMAGE

build stage: wheels → slim runtime

Python 3.12 slim, two stages

Non root user, uid 10001

Health check on /health/live

Knowledge files mounted, not baked in

COMPOSE SERVICES

db	pgvector/pgvector:pg16
api	the service on port 8000
ingest	one off job that loads the files

make up

CODEBASE IN ONE SLIDE

Read Files In, Verified Work Out

1 **Ingest**
parse, tag, chunk, embed, store

→

2 **Plan**
one search per purpose

→

3 **Retrieve**
hybrid, fused, reranked, packed

→

4 **Generate**
strict JSON, every item cited

→

5 **Verify**
check, repair, prune or refuse, audit

8,370 lines of typed Python in 12 packages, wired once in container.py, configured through environment and YAML.

09

Quality, Measured Not Assumed

Ninety six automated tests, a 24 question retrieval golden set, 13 generation scenarios and a human review gate before launch.

TEST PYRAMID

96 Tests, All Passing



Unit

Parsing, filters, fusion, planning, every check, pruning, OpenRouter contract, HTTP contract.

PostgreSQL

Migrations, filtered HNSW, full text, cascades, cache, audit.

Knowledge base

All files ingest; 12 activities per track tagged; reingestion idempotent.

SAFETY BEHAVIOUR, PROVEN BY TESTS

Nine Behaviours Locked In With Scripted Models

- ✓ Grounded task approved, cited, audited
- ✓ Hallucinated tool caught and repaired
- ✓ Invented complication pruned
- ✓ Unsupported core content refused
- ✓ No evidence, no model call
- ✓ Unavailable judge fails closed
- ✓ No dashes or banned terms reach candidates
- ✓ Personal data redacted before a model call
- ✓ Manager abstains when nothing answers

Source: docs/evaluation.md, section 1 | Quality | 168

RETRIEVAL GOLDEN SET

24 Questions With Known Right Answers

EXAMPLES FROM THE SET

- da_client_sponsor
- da_quality_log_deliverable
- da_store_codes
- da_activity_profiling
- programme_naming_convention

covering rules, all four tracks, brand, product, market evidence and methodology

Relevance by metadata

Right answers are defined by track, facet, section, company, activity number or text, so the set survives re chunking.

Four metrics at k = 5

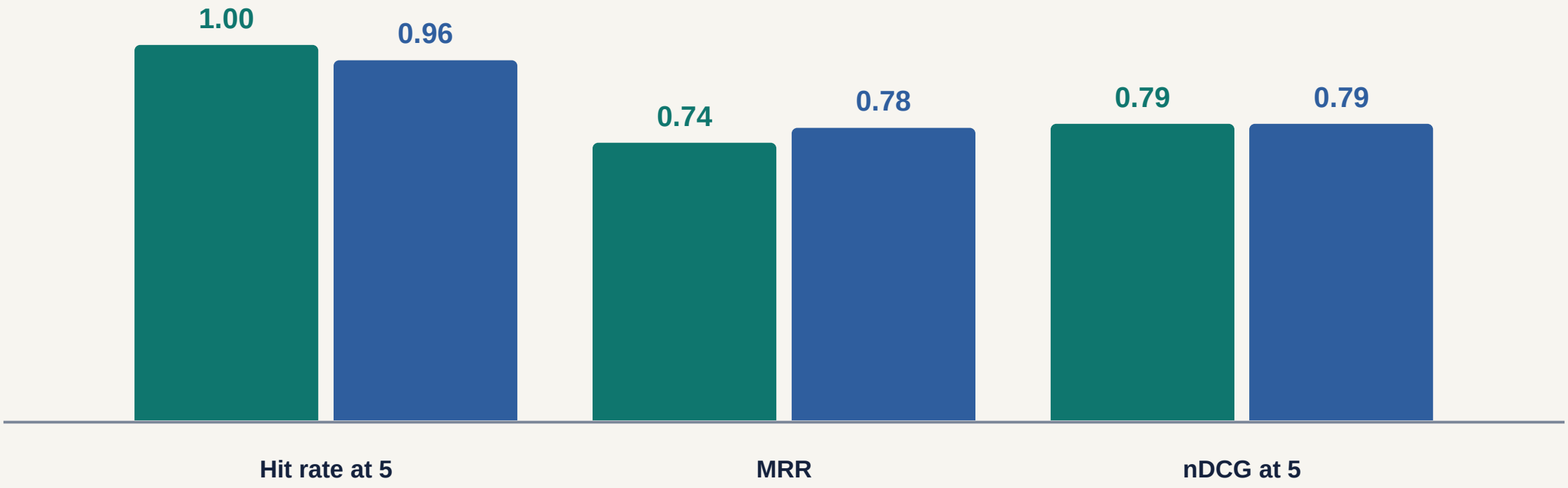
Hit rate, MRR, precision and nDCG.

Release blocker

A drop of more than 0.05 in hit rate or MRR.

RETRIEVAL RESULTS | OFFLINE BASELINE

Every Question Found Its Answer in Memory; 23 of 24 on PostgreSQL



■ in memory ■ PostgreSQL. Measured offline with the lexical stand in embedder; production embeddings should meet or exceed this.

GENERATION SCENARIOS

13 Scenarios, Including Three Designed to Say No

- da_activity_3
- da_activity_9_dashboard
- se_activity_5
- cy_activity_3_timeline
- de_activity_4
- cy_complication
- da_stakeholder_update
- se_colleague_request
- market_only_data_engineer
- nonexistent_activity: must abstain
- reply_deadline_question
- reply_unknown_fact: must abstain
- reply_wellbeing: must flag review

Targets: 100% behaving as expected; mean groundedness before pruning 0.95 or higher; zero rejections; p95 latency within 120 seconds.

END TO END RUN

The Full Pipeline, Exercised on the Real Knowledge Base

DATA ANALYST, ACTIVITY 3

Approved

groundedness 1.0

7 checks run

CYBERSECURITY, ACTIVITY 3

Approved

groundedness 1.0

7 checks run

WHAT WAS REAL, WHAT WAS MOCKED

Real: ingestion of all 27 files, planning, hybrid retrieval, evidence packs, every check, pruning, audit.

Mocked: the OpenRouter responses, because no live key was available in the build environment.

Next proof point: the same run with a live OpenRouter key, recorded as the production baseline.

STATIC QUALITY

Clean on Every Automated Gate

0	0	96	0
ruff findings	**mypy errors**	**tests passing**	**dashes in code text**
lint and format	type checks	unit and integration	style rule as a test

CI: GitHub Actions runs lint, types and tests on every push and pull request.

HUMAN REVIEW GATE

Machines Check Truth. People Check Believability.



At least 3 of 5 external testers

must describe the AI Manager as believable. Release gate from the Plain English Report.

THREE QUESTIONS PER TASK

- Would a real supervisor in this discipline write this?
- Is anything in it untrue to the brief?
- Is the tone right for a junior professional?

Findings recorded against audit ids.

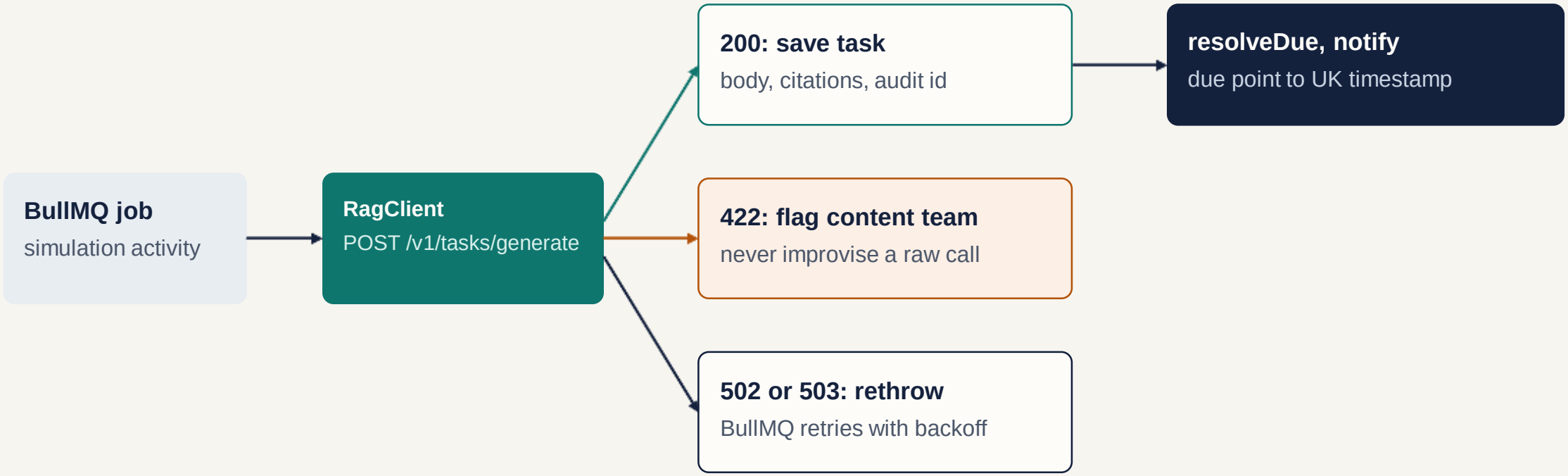
09

Integration and Operations

How the platform developer connects the Task Engine, how errors are handled, how the service runs on Google Cloud, and what to do when something goes wrong.

TASK ENGINE INTEGRATION

Generate Inside the Job, Then Notify the Candidate



ERROR CONTRACT

Every Status Code Has One Correct Reaction

Status	error code	Meaning	Platform should
200	none	verified output	save and deliver
401, 403	auth	wrong key	alert; do not retry
422	insufficient_evidence	knowledge gap	flag the content team
422	grounding_failed	output refused	retry later or flag
502	upstream_model_error	OpenRouter problem	retry with backoff
502	invalid_model_output	JSON unusable twice	retry with backoff
503	verification_unavailable	judge down	queue and retry

Every error body has the same shape: error, message, details. Details include the audit id and the verification report where relevant.

A CANDIDATE’S WEEK THROUGH THE SERVICE

Which Endpoint Serves Which Moment

MON	TUE	WED	THU	FRI
tasks: Activities 1 to 3	tasks: next activities	task: complication	task: colleague request	tasks: final activities
reply: client questions	reply: daily update feedback	reply: midweek review	reply: peer review questions	verify: evaluator feedback
	task: stakeholder update			

■ /v1/tasks/generate ■ /v1/manager/reply ■ special task types ■ /v1/verify for other flows. An illustrative schedule; the Task Engine decides timing.

KNOWLEDGE GRAPH MAPPING

Each Response Fills the Graph’s Task Node



title	task.title
deadline_at	resolveDue(task.schedule.due, week start)
priority	task.priority, P0 to P2, same enumeration
ASSIGNED_BY	the persona, passed as manager_name
evidence trail	evidence chunk ids and the audit id
task_id, assigned_at	set by the platform

DEPLOYMENT ON GOOGLE CLOUD

Matching the Platform’s Existing Footprint

PRIVATE NETWORK, PLATFORM INGRESS ONLY

Cloud Run service
API, minimum 1 instance, 1 vCPU, 1 GiB

AlloyDB
PostgreSQL with vector, 2 vCPU, 8 GiB

Cloud Run job
same image, command ingest

Secret Manager
OpenRouter key, API keys, database password

Cloud Storage
private bucket, files mounted read only

Monitoring
metrics scrape, JSON logs

Supabase also qualifies as the database; see the decision on slide 186.

Source: docs/operations.md, section 1 | Operations | 180

MONITORING

Four Alerts Worth Waking Up For

Rejected share above 5% in an hour

Prompt, model or knowledge regression. Inspect recent audit rows.

Any verification_unavailable

Judge model or OpenRouter degraded.

Abstentions rising after a content change

A brief section may have been renamed or removed.

/health/ready failing

Database, knowledge base, key or embedding mismatch.

Prompts and candidate messages are never logged; they live in the audit table, where retention is controlled.

SECURITY AND DATA PROTECTION

Defence in Depth, Built for UK GDPR

Network

Private ingress from the platform only.

Identity

Separate service and admin keys, constant time compare, rotation without downtime.

Personal data

Redacted before any model call.

Providers

OpenRouter routing with data collection denied.

Uploads

PDF, DOCX, XLSX only; 50 MB cap; base name only.

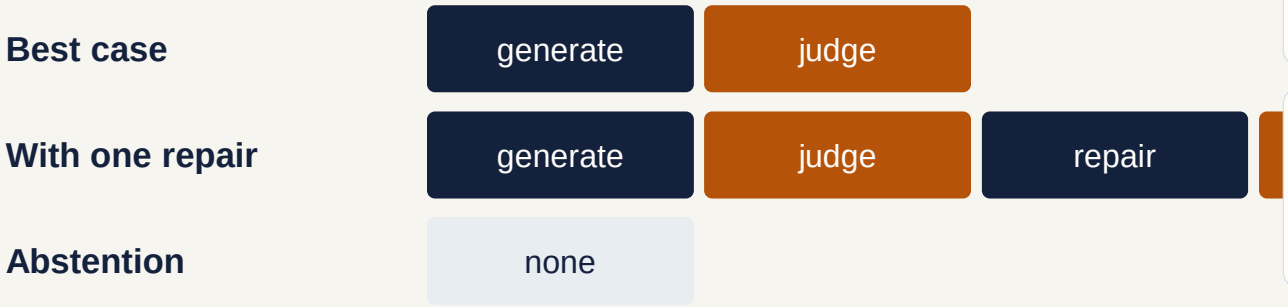
Runtime

Non root container; knowledge mounted read only.

The Work Brief is Programme Confidential: raw files are excluded from version control.

COST AND LATENCY

Bounded by Design: Two to Four Model Calls



Plus small query embeddings, cached; document embeddings paid once per content change.

Cheaper judge

Haiku class model for verification.

Off the critical path

Tasks are generated in background jobs; recommended timeout 120 s.

Measure, do not guess

Token counter per purpose gives the real bill after the first live week.

RUNBOOKS

Three Incidents, Three Written Answers

Many insufficient_evidence

1 Read details.missing

2 Check counts with stats

3 Check headings against tracks.yaml

4 Reingest with force, rerun eval

Rising grounding_failed

1 Group recent rejections by check

2 Entity or tool: check model, lower temperature

3 Entailment on message: review prompt version

4 Run eval_generation before and after

OpenRouter degraded

Retries and fallbacks are automatic.

Add more fallback models if needed.

Never disable verification to restore throughput. Queue and retry.

10

Decisions, Roadmap and Close

Four decisions needed from leadership, honest limitations, the next ninety days, risks, and what each audience should take away.

DECISIONS REQUESTED

Four Choices Only Leadership Can Make

1 Brand spelling

The brief says Vet2Employ; the brand portfolio says Vet2Mploy. One setting, V2E_BRAND_NAME, controls every output.

2 Supervisor persona names

The service never invents names. The platform should supply each persona’s name as manager_name.

3 Supabase or AlloyDB

Both work unchanged. AlloyDB matches the Google Cloud footprint; Supabase may already hold platform data.

4 UK only model regions

Whether OpenRouter traffic must be restricted to specific providers or regions for data residency.

HONEST LIMITATIONS

What Has Not Been Proven Yet

! **No live OpenRouter run yet.** Generation was verified with scripted model responses; the first live run sets the real baseline.

! **Retrieval numbers are offline.** Measured with a lexical stand in embedder, not text embedding 3 small.

! **Docker image not built here.** The Dockerfile and compose file are written and reviewed, not yet built in CI.

! **No human review yet.** The 3 of 5 believability gate has not been run.

ROADMAP

The Next 90 Days

DAYS 1 TO 30 | PROVE

Deploy to staging on Cloud Run

Ingest with production embeddings

Live eval: retrieval and 13 scenarios

Wire the Task Engine worker

DAYS 31 TO 60 | PILOT

Human review with 5 testers

One pilot cohort, one track

Dashboards and alerts live

Fill the three market gaps

DAYS 61 TO 90 | SCALE

All four tracks in production

Cost per candidate measured

Curated entity tables, first graph step

Evaluator grounding through /v1/verify

RISK REGISTER

Five Risks and How Each Is Contained

Risk	Likelihood	Impact	Mitigation
Too many refusals	Medium	Medium	Abstention metric by step; fix content, not the gate
Model provider outage	Medium	Medium	Fallback models, retries, queued jobs
Tasks feel templated	Medium	High	Market realism steps, human review gate, prompt iteration
Brief edited, headings change	Medium	Medium	Regex rules in YAML, knowledge base tests, eval after ingest
A subtle untruth slips through	Low	High	Seven layers, audit trail, review by audit id

FOR NON TECHNICAL STAKEHOLDERS

Five Things to Remember

1. The AI Manager only says what our documents say.
2. A second AI checks every sentence before a candidate sees it.
3. When unsure, it says so politely and asks a human.
4. Realism comes from 521 real UK job roles, not imagination.
5. Everything is recorded, so any answer can be traced and defended.

FOR TECHNICAL STAKEHOLDERS

Five Things to Know Before You Touch It

Plans are code

New request type: add a planner method and a schema.

Knowledge is config

New document: one manifest entry, then ingest.

Never bypass verification

Use /v1/retrieve and /v1/verify for direct OpenRouter flows.

Bump the prompt version

Any wording change, and rerun both evals.

Embedding change is a migration

New database, reingest, compare, switch.

Start here

make up, make test

GLOSSARY, PART 1

Terms Used in This Deck

RAG
Retrieval augmented generation: look up, then write.

Embedding
A list of numbers capturing meaning.

pgvector, HNSW
PostgreSQL extension and its fast vector index.

RRF
Reciprocal rank fusion: merging ranked lists.

Facet
The kind of fact a chunk holds.

Chunk
A passage of up to 450 tokens with metadata.

Token
Roughly three quarters of a word.

BM25, full text search
Keyword scoring that rewards rare exact words.

Rerank
Reordering candidates with extra signals.

Authority
Trust level of a source, 1 highest.

GLOSSARY, PART 2

Terms Used in This Deck

Evidence pack
The numbered passages E1 to En a model may use.

Hallucination
A fluent statement with no support.

Abstain
Decline to answer rather than guess.

OpenRouter
One API in front of many model providers.

Audit trail
One stored record per generation.

Grounded
Stated in, or directly implied by, the evidence.

Entailment
Whether a passage supports a claim.

Groundedness
Share of factual claims that are supported.

MRR, nDCG
Ranking quality measures between 0 and 1.

Fail closed
On doubt, refuse rather than pass.

THANK YOU

Questions and Discussion

“Every sentence a candidate reads can be traced to a source, or it is not sent.”

Vet2Employ RAG Service

Version 1.0.0, October 2026